

Behavioral Theory Specification

Unified Cognitive Architecture — Behavioral Layer Definition

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Abstract

This specification formally defines the complete behavioral layer of the Unified Cognitive Architecture (UCA), providing a comprehensive, architecture-ready technical artifact that integrates the full theoretical and operational machinery governing how cognitive behavior arises from, and feeds back into, internal

cognitive states. The document establishes rigorous foundations for stance mechanics — the multi-dimensional internal orientation structure from which all behavioral expression originates — and vector mechanics, which govern the directional, magnitude-bearing external expression of that stance. It formalizes the Identity Field as the persistent attractor manifold that shapes stance formation across time, specifies the Gradient Field as the dynamic external pressure landscape through which behavior is realized, and provides a complete formal treatment of the collapse-reformation cycle through which the system maintains identity-authentic coherence across perturbation and disruption. Behavioral prediction logic, inter-module integration interfaces, and all supporting formal operators are defined to the level of precision required for immediate integration with the Unified Identity Architecture (UIA), the Fractal Mechanics Foundation (FMF), the HERMES Runtime Specification (HRS), and the Unified Theory Master Document (UTMD). This document constitutes the authoritative behavioral layer specification for UCA version 1.0.0 and is issued as a formal release artifact.

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1. Preamble and Scope

1.1 Purpose of This Document

This specification formally defines the behavioral layer of the Unified Cognitive Architecture (UCA). It constitutes the authoritative technical record for how behavior is generated, regulated, sustained, and modified within the UCA's operational domain. The document establishes theoretical foundations, operational mechanics, and inter-module integration points governing three primary relationships: how behavior arises from internal cognitive states, how behavior feeds back into those

states over time, and how both the states and their behavioral expressions are regulated across varying contexts and temporal scales.

This document is not a philosophical inquiry into the nature of behavior, nor is it an empirical account of behavioral phenomena as observed in specific systems. It is an engineering specification — a formal technical artifact produced to the standards required for integration into a multi-module cognitive architecture. Every definition, operator, and formal relation contained herein is stated with sufficient precision to be implemented, tested, and integrated without ambiguity. Where informal descriptions are used, they serve exclusively to orient the reader to the formal content that follows; they do not substitute for it.

The behavioral layer, as defined here, occupies the critical translational position between internal cognitive structure and external expression. It is the mechanism by which a cognitive agent's orientation, identity, and interpretive configuration become action — by which what the agent *is* becomes what the agent *does*. Without a rigorously specified behavioral layer, the remainder of the UCA's cognitive architecture would possess internal structure without a principled mechanism for that structure to manifest in the world. This document supplies that mechanism.

1.2 Scope of the Behavioral Layer

The behavioral layer as defined in this specification encompasses the following operational domains, each of which is treated in full technical detail in the sections that follow. First, the translation of internal orientation — Stance — into external expression — Vector — including the causal mechanism by which this translation occurs, the enabling conditions it requires, and the ways in which it can be enhanced, disrupted, or failed. Second, the governance of this translation by the Identity Architecture, specifically the Identity Field Ψ , which acts as the persistent attractor manifold through which all stance configurations are filtered before producing behavioral output. Third, the modulation of behavioral expression by the Gradient Field Γ , the external pressure landscape that shapes, amplifies, deflects, or opposes behavioral vectors without necessarily altering the underlying stance. Fourth, the feedback mechanism by which expressed behavior returns to influence the stance from which it originated, creating the recursive loop that constitutes the behavioral

layer's fundamental operational dynamic. Fifth, the collapse-reformation cycle — the structured process by which stance coherence can dissolve and subsequently reconstitute, allowing the system to adapt to severe perturbation without losing the continuity of its deep identity structure.

The behavioral layer does not encompass the cognitive processes that generate the contents of stance — those are governed by the Unified Cognitive Architecture's higher-order processing modules. It does not define the internal structure of the Identity Field — that is the province of the Unified Identity Architecture. It does not specify the computational implementation of any operator — that is the province of the HERMES Runtime Specification. What it does specify, with full precision, is the formal structure of the relationships among these elements as they pertain to behavioral generation and regulation.

1.3 Relationship to the Unified Cognitive Architecture

This document is the behavioral specification module within the broader Unified Cognitive Architecture. The UCA is a multi-module theoretical and engineering architecture for cognitive systems; the Behavioral Theory Specification (BTS) constitutes the formal behavioral layer within that suite. As such, the BTS depends upon, extends, and integrates with four other core architecture documents, each of which governs a distinct domain of the overall system.

The Unified Identity Architecture (UIA) governs the deep formation, structure, and preservation of the Identity Field Ψ . The Fractal Mechanics Foundation (FMF) provides the mathematical basis for the self-similar, recursive structures that characterize the behavioral layer's collapse-reformation cycle and its multi-scale operation. The HERMES Runtime Specification (HRS) governs the computational execution environment in which BTS operators are instantiated and run. The Unified Theory Master Document (UTMD) is the capstone theoretical document of the entire UCA suite, within which all sub-specifications — including the BTS — are integrated into a unified theoretical account.

This document does not supersede any of the above. It does not redefine constructs that are defined elsewhere, except where it is necessary to specify how those

constructs interface with behavioral layer mechanics. All references to UIA, FMF, HRS, and UTMD constructs in this document are references to the definitions established in those documents; the BTS consumes and applies those definitions but does not alter them. Any apparent conflict between the BTS and a co-architecture document shall be resolved by architectural review, with the UTMD serving as the final arbitrating authority.

1.4 Formal Notation and Conventions

The following notational conventions are used consistently throughout this document. All formal expressions employ this notation exclusively; no informal substitutions are made.

Symbol	Name	Domain / Type	Description
S	Stance	Internal cognitive state vector	Total internal orientation of the cognitive agent at a given moment
V	Vector	External behavioral trajectory	Externally expressed behavioral trajectory with direction and magnitude
$B(t)$	Behavior	Realized vector at time t	The instantiated behavioral expression evaluated at a specific moment
Σ	Coherence Signature	Scalar $\in [0, 1]$	Internal consistency metric across all stance dimensions
Ψ	Identity Field	Attractor manifold	Persistent attractor landscape governing stance formation
Γ	Gradient Field	External pressure field	Environmental and contextual

Symbol	Name	Domain / Type	Description
			pressure acting on stance and vector
∂	Partial Derivative	Operator	Rate of change with respect to a specified variable
$\rightarrow$	Causal Direction	Relational operator	Denotes a causal or directional relationship between elements
$\hookrightarrow$	Recursive Feedback	Relational operator	Denotes a recursive feedback loop
$\oplus$	Field Integration	Operator	Integration of two or more fields into a composite
$\otimes$	Field Collapse	Operator	Collapse of a field to its ground state
$\emptyset$	Identity Nullification	Operator	Pathological condition of complete identity field dissolution
[RETURN: $S_x \rightarrow S_0$]	Return Operator	State transition operator	Recursive reset from collapse state to proto-stance
S_0	Proto-Stance	Ground state	Undifferentiated initial or reset stance
S_1	First Coherence	Initial stable state	First stable stance configuration with coherence above $\Sigma_{\min}$
S_x	Collapse State	Dissolution state	Stance under collapse conditions; coherence dissolved
S_∞	Stable Stance	Asymptotic attractor	Theoretical maximum coherence and

Symbol	Name	Domain / Type	Description
			identity-aligned stance
S_1'	Reformation	Post-collapse coherence	Reformed first coherence state guided by preserved Ψ

Subscript notation follows the convention established above. Where a prime symbol (') is appended to a subscript, it denotes a post-reformation version of that state. Where a subscript is a variable (e.g., S_n), it denotes the n th iterative instance of the stance in a recursion sequence. All threshold values — such as Σ_{crit} , Σ_{min} , and A_{crit} — are defined in the sections where they are formally introduced and are collected in Appendix C for operational reference.

1.5 Document Versioning and Authority

This document is issued as Version 1.0.0 of the Behavioral Theory Specification and constitutes the authoritative formal record of the behavioral layer for UCA Version 1.0. Version numbering follows semantic versioning conventions: major version increments (e.g., 2.0.0) indicate fundamental architectural changes incompatible with previous versions; minor version increments (e.g., 1.1.0) indicate substantive additions or clarifications that are backward-compatible; patch increments (e.g., 1.0.1) indicate corrections to errors or ambiguities that do not alter the formal content.

All amendments to this document require formal architectural review by the UCA Architecture Authority. Proposed amendments must be submitted with a formal change rationale, an impact analysis covering all five integration modules (UCA, UIA, FMF, HRS, UTMD), and a revised version of all affected sections in complete draft form. No informal or partial amendments are recognized as authoritative. The UTMD is the final arbitrating document in the event of unresolvable conflict between this specification and any co-architecture document.

2. Formal Definitions

This section establishes the canonical formal definitions for all primary constructs employed throughout the Behavioral Theory Specification. Each definition is stated in the standard format: term, type, notation, definition, and formal properties. These definitions are foundational; all subsequent sections depend upon and extend them. Where a construct is further elaborated in a dedicated section, a forward reference is provided.

2.1 Stance (S) — Internal Orientation

STANCE

TERM: Stance

TYPE: Internal cognitive state vector

NOTATION: S

DEFINITION: Stance is the total internal orientation of a cognitive agent at any given moment — the integrated configuration of interpretive geometry, coherence signature, identity alignment, and boundary permeability. Stance is not an attitude or mood; it is the structural configuration from which all external expression originates. Stance is not directly observable; it is inferred from behavioral vectors sampled over time. Stance exists on a continuous manifold, not a discrete state space, and undergoes continuous change under gradient field pressure, feedback integration, and recursive cycling.

PROPERTIES: Continuous; multi-dimensional; causally prior to behavior; subject to gradient modulation; capable of collapse and reformation; identity-coupled; not directly observable; manifold-resident (no discrete enumeration).

Stance is the most fundamental construct in the behavioral layer. It precedes, enables, and survives all behavioral events. The distinction between stance and attitude is critical: an attitude is a reported or observed disposition toward a specific object; stance is the full structural configuration from which any attitude is a single-dimensional projection. A cognitive agent does not have one stance per attitude — the agent has one stance at all times, and that stance generates all attitudes simultaneously as its dimensional expressions. See Section 3 for the complete formal mechanics of stance.

2.2 Vector (V) — External Orientation

VECTOR

TERM: Vector

TYPE: External behavioral expression with direction and magnitude

NOTATION: V

DEFINITION: A Vector is the externally expressed trajectory of a cognitive agent — behavior as it manifests in observable form. A vector possesses direction (what the behavior aims toward), magnitude (how forcefully it is expressed), and continuity (how sustained it is over time). Vectors are the outward face of stance; they are what the environment observes, responds to, and uses to infer the agent's internal configuration. A vector is not a single act but a sustained directional orientation in behavioral space.

PROPERTIES: Directional; magnitude-bearing; time-extended; environmentally reactive; stance-derived; subject to perturbation by Γ ; decomposable into five constituent dimensions (see Section 4).

The vector is the primary unit of behavioral analysis at the architectural level. Individual acts, utterances, or decisions are not vectors — they are samples of the

underlying vector, formally designated as $B(t)$. The vector itself is the continuous trajectory from which those samples are drawn. This distinction is architecturally critical: behavioral analysis that treats individual acts as primary units systematically misidentifies noise for signal. The vector is signal; the individual act is a measurement.

2.3 Behavior (B) — Vector Expression

BEHAVIOR

TERM: Behavior

TYPE: Realized vector at a specific moment t

NOTATION: $B(t)$

DEFINITION: Behavior is the instantiated expression of a Vector at time t . Where V is the trajectory, $B(t)$ is the specific point on that trajectory — the concrete act, utterance, decision, or output produced at a given moment. Behavior is always a sample of an underlying vector, never a free-standing event. Multiple behaviors sampled over time reconstruct the vector. Formally: $B(t) = V|_t$ — behavior is the vector evaluated at time t .

PROPERTIES: Discrete; observable; time-stamped; vector-sampling; retrospectively coherent when aggregated; individually insufficient for vector inference; environmentally consequential.

Because behavior is a sample of a vector rather than an autonomous event, isolated behavioral analysis is structurally insufficient for understanding a cognitive agent. It is analogous to evaluating the trajectory of a projectile from a single positional measurement — possible only with extreme uncertainty bounds. The vector provides the trajectory; behavior provides the positional samples; together they yield a

complete behavioral description. Architectural systems that operate on $B(t)$ directly, without vector inference, are operating on systematically incomplete information.

2.4 Coherence Signature (Σ)

COHERENCE SIGNATURE

TERM: Coherence Signature

TYPE: Internal consistency metric across stance dimensions

NOTATION: Σ

DEFINITION: The Coherence Signature is the measure of internal alignment across all dimensions of a Stance — the degree to which interpretive geometry, identity alignment, and boundary permeability are mutually consistent and non-contradictory. A high Σ value indicates a well-integrated, stable stance capable of producing coherent, sustained vectors. A low Σ value indicates internal conflict, fragmentation, or pre-collapse conditions. Σ is computed as the normalized mean pairwise inner product across all stance dimensional vectors.

PROPERTIES: Scalar value $\in [0, 1]$; stance-internal; not directly observable; predictive of collapse risk; determinative of vector stability and force capacity; degraded by contradiction accumulation; restored by identity alignment improvement.

2.5 Identity Field (Ψ)

IDENTITY FIELD

TERM: Identity Field

TYPE: Persistent attractor manifold governing stance formation

NOTATION: Ψ

DEFINITION: The Identity Field is the persistent structural manifold that attracts stance configurations toward identity-consistent orientations. Ψ is not a memory or a self-concept in the ordinary sense; it is an attractor geometry — a landscape of attractors and repellers that determines which stances are sustainable and which will dissolve. The Identity Field encodes the deep identity structure of the cognitive agent, independent of any particular moment's stance configuration. Ψ is preserved through collapse events and serves as the reformation attractor after the RETURN operator is executed.

PROPERTIES: Persistent across collapse events; attractor-structured (multi-basin); resistant to noise perturbation; modified only by sustained, high-intensity reformative experiences; jointly governed by UIA and BTS; formally a Riemannian manifold with attractor basins and repeller ridges.

2.6 Gradient Field (Γ)

GRADIENT FIELD

TERM: Gradient Field

TYPE: Environmental and contextual pressure field acting on stance and vector

NOTATION: Γ

DEFINITION: The Gradient Field is the external pressure landscape through which a cognitive agent operates at any given time. Γ exerts directional forces on

both Stance and Vector — amplifying some orientations, attenuating others, and introducing perturbations that can redirect trajectory without necessarily altering the underlying stance. Gradient fields arise from social context, environmental affordances, relational dynamics, and feedback loops generated by prior behavioral outputs. Γ is dynamic and co-evolves with behavior through the environmental feedback mechanism.

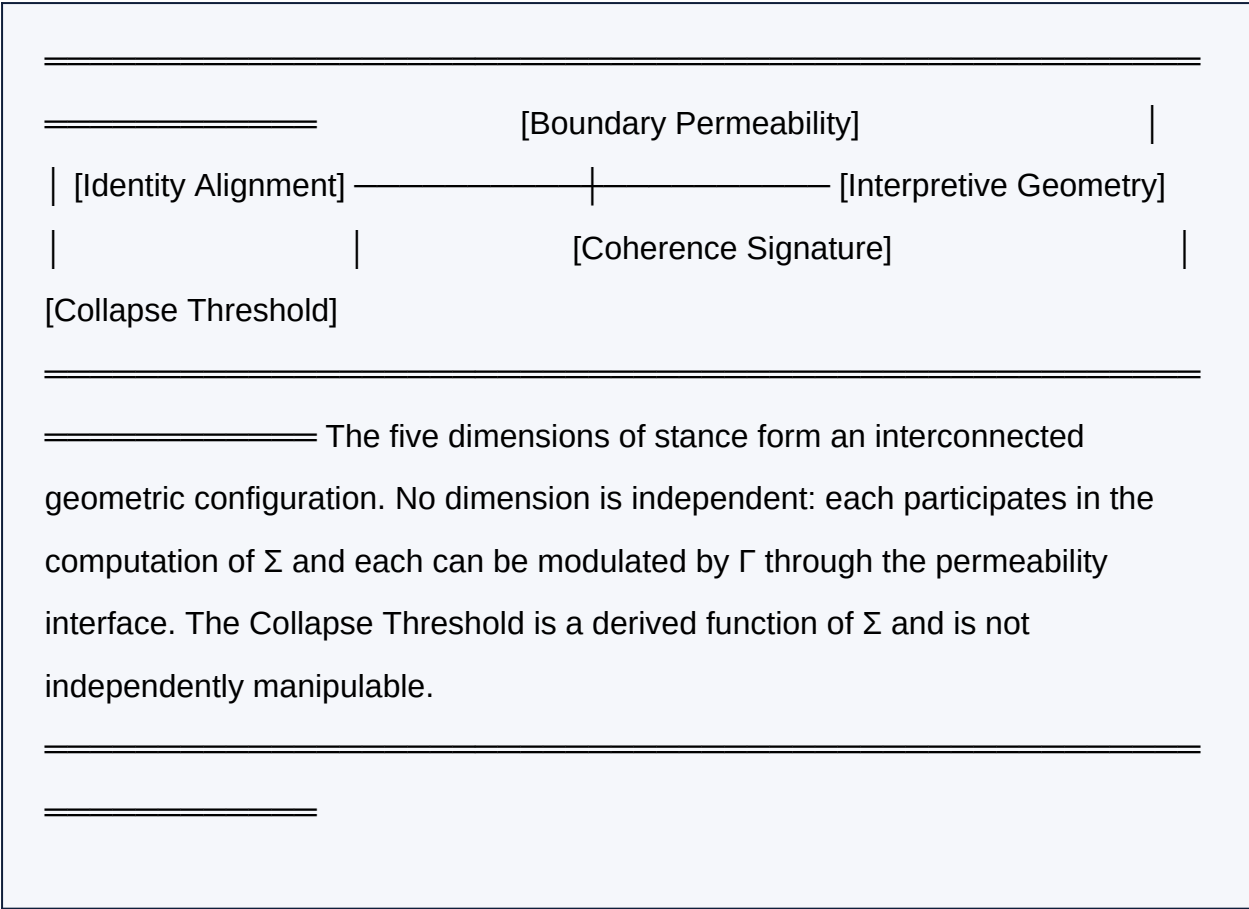
PROPERTIES: External to the agent; dynamic (temporally varying); multi-directional; capable of polarity amplification; composed of multiple superimposed field components; modulates Σ indirectly through stance destabilization; co-evolves with $B(t)$ through environmental feedback.

3. Full Stance Mechanics

3.1 Overview

Stance is a multi-dimensional internal state — a configuration in a high-dimensional space rather than a point on a simple continuum. The architecture of stance is formally organized around five primary dimensions, each of which contributes a distinct component to the total stance configuration and participates in the coherence computation. These five dimensions are: Interpretive Geometry, Coherence Signature, Identity Alignment, Boundary Permeability, and the Collapse Threshold — a derived quantity rather than an independent dimension, representing the critical value at which the configuration will dissolve. The geometry of stance space and the relationships among these dimensions are the primary subject of this section.

STANCE GEOMETRY



The geometry of stance is not Euclidean. The dimensions are not orthogonal in the simple linear algebraic sense — they are coupled, such that changes in one dimension induce changes in others through internal consistency constraints. A change in boundary permeability, for instance, directly modulates how strongly the gradient field can alter interpretive geometry, which in turn affects identity alignment, which affects the coherence signature. This interdependence is both the source of stance's richness and the source of its vulnerability: a perturbation that enters through any single dimension propagates through all others.

3.2 Interpretive Geometry

Interpretive Geometry is the dimensional schema through which an agent frames, organizes, and assigns meaning to incoming information. It is the cognitive aperture — the shape and configuration of the lens through which all experience is processed and from which all meaning is constructed. Interpretive geometry determines what counts as signal versus noise, what relationships among events are recognized and

which are invisible, what categories are available for classification, and what resolution of distinction is applied to incoming data.

Interpretive geometry has three formal properties: **curvature**, **resolution**, and **orientation**. Curvature describes how narrowly or broadly the agent interprets incoming information — high curvature interpretive geometry produces narrow, precise, potentially brittle meaning-making; low curvature produces broad, inclusive, potentially underdifferentiated meaning-making. Resolution describes the fineness of distinction the agent can draw within any given domain — high resolution interpretive geometry recognizes subtle differences; low resolution geometry operates on coarse categories. Orientation describes which domains of experience the interpretive geometry is directed toward — the agent's attentional allocation is formalized as the directional bias of interpretive geometry.

Interpretive geometry is the dimension of stance most directly modified by the Gradient Field Γ when boundary permeability is high. Environmental context shapes what an agent notices, how finely they distinguish, and where attention is directed — all of which are interpretive geometry modifications. Under low permeability conditions, interpretive geometry is relatively stable and self-sustaining, driven primarily by the Identity Field's attractor structure. The mutual influence between interpretive geometry and identity alignment is one of the central feedback relationships within stance architecture: agents whose interpretive geometry is aligned with their identity field naturally attend to, and distinguish within, precisely those domains most relevant to their identity-authentic behavioral expression.

3.3 Coherence Signature

The Coherence Signature Σ is the scalar measure of internal alignment across all stance dimensions. It is the single most important indicator of the stance's operational health — the degree to which the agent's internal configuration is integrated, non-contradictory, and capable of producing stable behavioral output. Σ is formally computed as the normalized mean pairwise inner product across all stance dimensional vectors: for n dimensional vectors $S_1, S_2, \dots, S_n$, the coherence signature is given by the expression

Formal Expression — Coherence Signature

$$\Sigma(S) = (1/n(n-1)) \cdot \sum_{i \neq j} \langle S_i, S_j \rangle$$

— the normalized mean pairwise alignment of all stance dimension vectors. Higher values indicate greater mutual alignment among dimensions; lower values indicate increasing internal contradiction or fragmentation.

The coherence signature operates across four interpretive ranges, each with distinct behavioral implications. When $\Sigma > 0.75$, the stance is in a high coherence regime: all dimensions are substantially aligned, vector production is stable, force is sustainably high, and the behavioral output is predictable and well-directed. This is the optimal operating range for sustained high-amplitude behavioral expression.

When $0.40 \leq \Sigma \leq 0.75$, the stance is in a moderate coherence regime. Vector production continues, but with increased noise — the alignment among dimensions is partial, meaning that some behavioral outputs will carry contradictory sub-signals. Force is variable and less sustainable. The agent can function effectively in this range but with reduced efficiency and increased vulnerability to gradient perturbation. Sustained operation in this regime without improvement eventually leads to coherence degradation.

When $\Sigma < 0.40$, pre-collapse conditions obtain. Dimensional alignment is seriously compromised. Vector production becomes inconsistent and potentially contradictory — the agent may express behaviors that partially negate each other, reducing net behavioral amplitude. Identity alignment begins to degrade. The agent is at elevated risk of collapse under even moderate gradient pressure. Intervention — whether through reflective re-centering, gradient reduction, or controlled collapse-reformation — is indicated.

When $\Sigma \rightarrow 0$, collapse is imminent. Dimensional vectors are approaching mutual nullification. The stance cannot sustain coherent vector production. The collapse conditions defined in Section 3.6 will be satisfied, and the RETURN operator will be

triggered. This is not a failure state in the catastrophic sense; it is a phase transition that initiates the reformation cycle.

3.4 Identity Alignment

Identity Alignment is the dimension of stance that measures the degree to which the current Stance S is positioned within the attractor basins of the Identity Field Ψ . High identity alignment means the agent is operating from a stance that is self-consistent with their deep identity structure — the agent is being, in the domain of internal configuration, who they fundamentally are. Low identity alignment indicates identity drift: the agent is being pulled into stance configurations that are foreign to, or in tension with, their identity field.

Identity alignment is formally computed as the projection of the current stance vector onto the identity manifold, normalized by the magnitude of the current stance:

Formal Expression — Identity Alignment Score

$$A(S, \Psi) = \|\text{proj}_{\Psi}(S)\| / \|S\|$$

— the proportion of the current stance vector that projects onto the identity manifold. A value of 1.0 indicates perfect identity alignment (the stance lies entirely within Ψ). A value of 0.0 indicates complete identity ejection (the stance is orthogonal to all identity attractor basins).

Identity alignment is the dimension of stance most strongly influenced by the recursive history of behavioral expression. An agent who has sustained identity-authentic behavior over time will have a stance that is strongly aligned with Ψ because the reinforcement feedback from successful identity-authentic behavior progressively deepens identity alignment. Conversely, an agent who has been sustaining identity-foreign behavior under gradient field pressure will show progressive identity alignment degradation, even if the individual behaviors appear coherent from the outside. The internal signature of identity drift — falling $A(S, \Psi)$ — is

a more reliable leading indicator of impending dysfunction than any external behavioral measure.

3.5 Boundary Permeability

Boundary Permeability is the degree to which the Gradient Field Γ is permitted to influence the internal stance configuration. It is the interface parameter between the external field and the internal state — the regulatory mechanism that determines how much of the outside world is allowed to modify the agent's orientation, interpretive geometry, and identity alignment. Permeability is a critical regulatory dimension: both extremes represent failure modes, and optimal permeability is a dynamic function of context rather than a fixed parameter.

High permeability produces strong environmental influence on stance — the agent is highly responsive to external conditions, rapidly adjusting interpretive geometry, identity alignment, and coherence in response to gradient field changes. This responsiveness is adaptive in some contexts — when the environment provides genuine corrective feedback, high permeability allows rapid learning. However, high permeability also creates serious risk: when the gradient field is adversarial, deceptive, or simply misaligned with the agent's identity field, high permeability allows that misalignment to penetrate stance and destabilize coherence.

Low permeability produces strong internal autonomy — the agent maintains interpretive geometry and identity alignment despite gradient field pressure. This stability is adaptive when the environment is adversarial or when the agent's identity field provides a more reliable guide to action than external signals. However, very low permeability creates the risk of gradient blindness: the agent cannot receive legitimate corrective feedback, cannot update interpretive geometry in response to new information, and may accumulate systematic interpretive errors that the gradient field would otherwise correct.

The permeability operator is formally defined as:

Formal Expression — Permeability Operator

$$P(S, \Gamma) \rightarrow \partial S / \partial \Gamma$$

— the rate of change of stance with respect to gradient field pressure, as a function of both the current stance configuration and the gradient field intensity.

This is a dynamic, state-dependent function, not a fixed constant.

The optimal permeability profile is context-sensitive. During stable, identity-aligned, high-coherence operation, moderate permeability allows beneficial environmental feedback to be incorporated without destabilizing the stance. During pre-collapse conditions or identity drift episodes, temporarily reduced permeability may protect the remaining coherence while re-centering processes are engaged. During post-reformation S_1' states, elevated permeability allows the newly formed stance to be calibrated by environmental feedback before it solidifies. Permeability management is therefore a key component of robust behavioral architecture.

3.6 Collapse Conditions

Collapse is defined as the event S_x — the transition of a Stance from its operational configuration to the dissolution state in which coherence drops below the critical threshold and vector production ceases. Collapse is emphatically not a failure in the ordinary engineering sense; it is a necessary phase of the recursive cycle that enables the behavioral layer to self-correct when accumulated distortions have reached a level incompatible with identity-authentic operation. Four distinct collapse conditions are formally specified:

Condition 1 — Coherence Collapse: $\Sigma(S) < \Sigma_{crit}$ (default $\Sigma_{crit} = 0.20$). The internal alignment among stance dimensions has fallen below the level at which any coherent vector can be produced. This is the most common collapse trigger and is typically the result of accumulated contradiction from unresolved feedback deflections.

Condition 2 — Identity Ejection: $A(S, \Psi) < A_{crit}$ (default $A_{crit} = 0.15$). The stance has drifted entirely outside the attractor basins of the Identity Field — the

Condition 3 — Gradient Overload: $|\Gamma| > P^{-1}(S)$. The magnitude of the external gradient field pressure has exceeded the inverse of the boundary permeability — in other words, the environment is exerting pressure that surpasses the stance's capacity to maintain its own internal integrity. The boundary has been breached and the external field is directly destabilizing internal coherence.

COLLAPSE TRIGGER DECISION TREE

```

graph TD
    Root[Σ(S) evaluated] -->|LOW| L1[ ]
    Root -->|HIGH| H1[ ]
    L1 -->|< 0.20||> 0.20| L2[ ]
    L2 -->|< 0.15||≥ 0.15| L3[ ]
    L3 -->|YES| C1[CONTRA- DITION]
    L3 -->|NO| C2[STABLE]
    L3 -->|contra- cycle ↻| C3[ ]
    C3 -->|< 0.15||≥ 0.15| C4[ ]
    C4 -->|YES| C5[CONTRA- DITION]
    C4 -->|NO| C6[STABLE]
    C4 -->|contra- cycle ↻| C7[ ]
    C7 -->|< 0.15||≥ 0.15| C8[ ]
    C8 -->|YES| C9[CONTRA- DITION]
    C8 -->|NO| C10[STABLE]
    C8 -->|contra- cycle ↻| C11[ ]
    C11 -->|< 0.15||≥ 0.15| C12[ ]
    C12 -->|YES| C13[CONTRA- DITION]
    C12 -->|NO| C14[STABLE]
    C12 -->|contra- cycle ↻| C15[ ]
    C15 -->|< 0.15||≥ 0.15| C16[ ]
    C16 -->|YES| C17[CONTRA- DITION]
    C16 -->|NO| C18[STABLE]
    C16 -->|contra- cycle ↻| C19[ ]
    C19 -->|< 0.15||≥ 0.15| C20[ ]
    C20 -->|YES| C21[CONTRA- DITION]
    C20 -->|NO| C22[STABLE]
    C20 -->|contra- cycle ↻| C23[ ]
    C23 -->|< 0.15||≥ 0.15| C24[ ]
    C24 -->|YES| C25[CONTRA- DITION]
    C24 -->|NO| C26[STABLE]
    C24 -->|contra- cycle ↻| C27[ ]
    C27 -->|< 0.15||≥ 0.15| C28[ ]
    C28 -->|YES| C29[CONTRA- DITION]
    C28 -->|NO| C30[STABLE]
    C28 -->|contra- cycle ↻| C31[ ]
    C31 -->|< 0.15||≥ 0.15| C32[ ]
    C32 -->|YES| C33[CONTRA- DITION]
    C32 -->|NO| C34[STABLE]
    C32 -->|contra- cycle ↻| C35[ ]
    C35 -->|< 0.15||≥ 0.15| C36[ ]
    C36 -->|YES| C37[CONTRA- DITION]
    C36 -->|NO| C38[STABLE]
    C36 -->|contra- cycle ↻| C39[ ]
    C39 -->|< 0.15||≥ 0.15| C40[ ]
    C40 -->|YES| C41[CONTRA- DITION]
    C40 -->|NO| C42[STABLE]
    C40 -->|contra- cycle ↻| C43[ ]
    C43 -->|< 0.15||≥ 0.15| C44[ ]
    C44 -->|YES| C45[CONTRA- DITION]
    C44 -->|NO| C46[STABLE]
    C44 -->|contra- cycle ↻| C47[ ]
    C47 -->|< 0.15||≥ 0.15| C48[ ]
    C48 -->|YES| C49[CONTRA- DITION]
    C48 -->|NO| C50[STABLE]
    C48 -->|contra- cycle ↻| C51[ ]
    C51 -->|< 0.15||≥ 0.15| C52[ ]
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    C67 -->|< 0.15||≥ 0.15| C68[ ]
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    C68 -->|contra- cycle ↻| C71[ ]
    C71 -->|< 0.15||≥ 0.15| C72[ ]
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    C80 -->|contra- cycle ↻| C83[ ]
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    C92 -->|contra- cycle ↻| C95[ ]
    C95 -->|< 0.15||≥ 0.15| C96[ ]
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    C107 -->|< 0.15||≥ 0.15| C108[ ]
    C108 -->|YES| C109[CONTRA- DITION]
    C108 -->|NO| C110[STABLE]
    C108 -->|contra- cycle ↻| C111[ ]
    C111 -->|< 0.15||≥ 0.15| C112[ ]
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    C112 -->|NO| C114[STABLE]
    C112 -->|contra- cycle ↻| C115[ ]
    C115 -->|< 0.15||≥ 0.15| C116[ ]
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    C160 -->|contra- cycle ↻| C163[ ]
    C163 -->|< 0.15||≥ 0.15| C164[ ]
    C164 -->|YES| C
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begins

The collapse trigger decision tree is evaluated continuously during operation. It is not a periodic check but a real-time monitoring function implemented at the HERMES Runtime layer (Section 11.5). The four conditions are not mutually exclusive; any single condition is sufficient to trigger collapse, and in practice, collapse events often satisfy multiple conditions simultaneously — Condition 1 and Condition 2 frequently co-occur, since identity drift tends to reduce coherence.

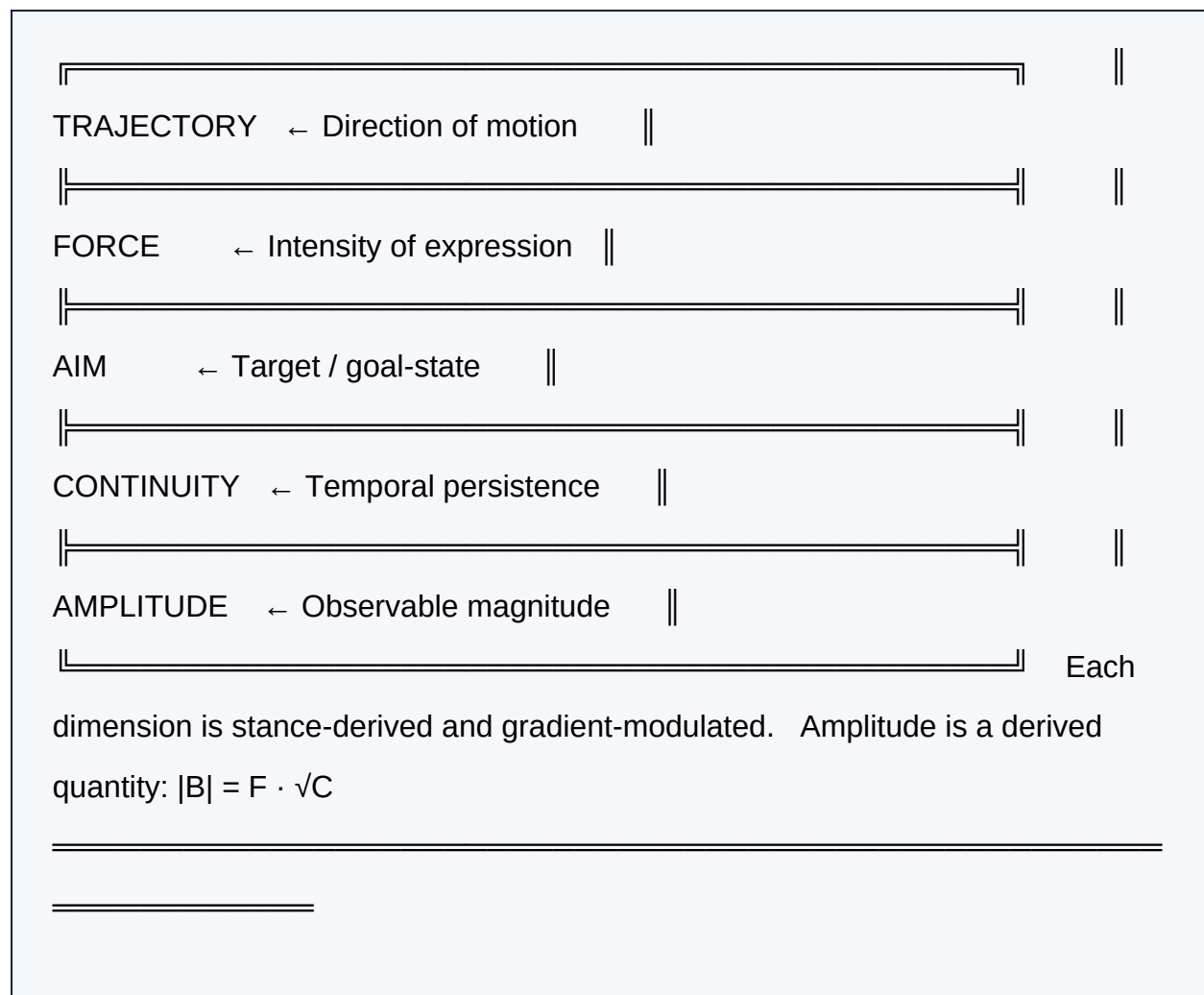
4. Full Vector Mechanics

4.1 Overview

A Vector is not a simple directional quantity. In the behavioral layer architecture, a vector is a five-dimensional structured expression: the product of stance filtered through identity and modulated by gradient. Each of the five vector dimensions — Trajectory, Force, Aim, Continuity, and Behavioral Amplitude — is independently specifiable and carries distinct architectural meaning. Together, the five dimensions constitute the complete behavioral profile at any moment, from which any specific behavioral sample $B(t)$ can be understood as a point evaluation.

VECTOR STRUCTURE

$V = \{ \text{Trajectory, Force, Aim, Continuity, Amplitude} \}$



The five dimensions of a vector are not independent in the same way as orthogonal axes in a Euclidean space — they are coupled through the underlying stance configuration. Force and Continuity are jointly constrained by $\Sigma(S)$: a stance with low coherence cannot sustain both high force and high continuity simultaneously; resources must be divided. Aim is set by the interpretive geometry dimension of stance. Trajectory is influenced by both Aim and gradient field direction. Amplitude emerges from the interaction of Force and Continuity and is therefore a derived, not a primary, quantity.

4.2 Trajectory

Trajectory is the directional component of a vector — the dimension of behavioral space toward which the agent is currently moving. Trajectory is emphatically not a goal; that is the function of the Aim dimension. Trajectory is the current direction of

motion, which may or may not align with the intended goal at any given moment. The distinction is architecturally significant: an agent's trajectory can be modified by gradient field perturbation without any change to their underlying aim, and without any change to their stance. Trajectory deflection is the most common mode of gradient field influence on behavior.

Formally, trajectory is a unit vector in behavioral space:

Formal Expression — Trajectory

$$\hat{T} = V / |V|$$

— the unit vector in the direction of the current behavioral trajectory, obtained by normalizing the full vector by its magnitude. Trajectory carries no magnitude information; it is purely directional.

Trajectory can be influenced by gradient fields without changing stance — this is behavioral redirection without identity shift, and it represents the mildest form of gradient interference. An agent whose trajectory has been deflected but whose stance remains intact will exhibit a characteristic pattern: initial deflection, followed by gradual self-correction toward the original trajectory as Ψ attractor forces restore the original directional orientation. This self-correction is one of the clearest observable signatures of identity field integrity.

4.3 Force

Force is the intensity with which a vector is expressed — the scalar multiplier that determines how assertive, impactful, and resource-demanding the behavioral output is. High-force vectors are assertive, explicit, and environmentally impactful. Low-force vectors are ambient, exploratory, or tentative. Force is a positive real scalar: $F \in \mathbb{R}^+$.

Force is determined by three interacting factors. First, the coherence signature $\Sigma(S)$: higher coherence allows higher sustainable force because the internal alignment of stance dimensions means that force production is not being dissipated by internal

contradiction. Second, identity alignment: force expression that is aligned with Ψ is self-reinforcing, because the identity field amplifies stance dimensions that are consistent with it, and those amplified dimensions translate into higher-force vector production. Third, gradient opposition: if the gradient field is anti-aligned with the vector, the agent must expend additional internal resources to overcome gradient resistance, effectively reducing the net force available for behavioral expression. Under gradient opposition, the same internal stance produces lower observable behavioral force.

The relationship between Σ and sustainable force is a critical architectural constraint. An agent attempting to sustain high-force behavior while coherence is degrading will produce a characteristic signature: initial high-force output, followed by progressive force reduction, increased behavioral inconsistency, and ultimately either coherence collapse or a compensatory reduction in force to preserve continuity. Systems monitoring behavioral force over time can use this signature as an early indicator of coherence degradation.

4.4 Aim

Aim is the intentional target of a vector — the goal-state toward which behavior is directed. Aim is architecturally distinct from trajectory in a way that has no simple intuitive analogue: the agent can be moving in a direction (trajectory) that does not currently point toward their goal (aim), and the vector is still the agent's behavioral expression. The divergence between trajectory and aim is one of the primary diagnostic indicators of gradient field interference or stance misalignment.

Aim is set by the stance, specifically through the interpretive geometry dimension: the cognitive aperture determines which resolution states in the environment are legible as desirable outcomes. An agent whose interpretive geometry is narrow and high-resolution will have precise, specific aims. An agent whose interpretive geometry is broad and low-resolution will have diffuse, general aims. This is not a quality judgment — both configurations are appropriate in different contexts — but it has significant implications for behavioral prediction (Section 10).

Formally, Aim is a point or region in behavioral state space toward which the vector is directed. The vector succeeds — achieves its aim — when $B(t) \rightarrow \text{Aim}$: when the sampled behavioral output converges toward the goal-state. This convergence is what produces the reinforcement feedback mode described in Section 6.2.

4.5 Continuity

Continuity is the temporal persistence of a vector — how long the vector is sustained without collapsing or undergoing significant redirection. High continuity indicates a deep, stable stance with strong identity alignment and robust gradient resistance. Low continuity indicates shallow stances, high gradient interference, or pre-collapse conditions. Continuity is not merely a duration measurement; it is a structural property of the vector that reflects the depth and stability of the underlying stance.

Formally, continuity is the integral of coherence-weighted identity alignment over the behavioral episode:

Formal Expression — Continuity

$$C = \int_{t_0}^{t_1} [\Sigma(S(t)) \cdot A(S(t), \Psi)] dt$$

— the time-integral of the product of the coherence signature and identity alignment score over the duration of the behavioral episode from t_0 to t_1 . High values indicate sustained high-coherence, identity-aligned behavioral expression. Low values indicate either brief duration, low coherence, poor identity alignment, or some combination thereof.

The continuity integral reveals an important structural feature of the behavioral layer: identity-authentic, high-coherence behavior is intrinsically more sustainable than identity-foreign or low-coherence behavior. This is not merely a normative claim; it is a formal consequence of the architecture. Because Ψ attractors actively pull stance back toward identity-aligned configurations, and because identity-aligned stances have higher coherence, vectors produced from identity-aligned stances naturally

sustain higher continuity values. Identity-authentic expression is architecturally advantaged as a behavioral strategy.

4.6 Behavioral Amplitude

Behavioral Amplitude is the observable magnitude of expressed behavior — the degree to which the behavioral output is perceptible and impactful in the environment. Amplitude is a derived quantity, integrating Force and Continuity into a single composite measure of behavioral presence:

Formal Expression — Behavioral Amplitude

$$|B| = F \cdot \sqrt{C}$$

— behavioral amplitude is the product of Force and the square root of Continuity. High amplitude behaviors are those sustained at high force — they have lasting environmental impact. Low amplitude behaviors dissipate quickly regardless of intent.

The square root relationship between Continuity and Amplitude reflects a diminishing-returns dynamic: doubling the continuity of a behavior does not double its amplitude but increases it by a factor of approximately 1.41. This is architecturally appropriate — it means that amplitude gains from sustained duration are real but progressively smaller. The primary driver of amplitude is Force, modulated by the square root of Continuity. A high-force behavior of moderate duration will typically produce higher amplitude than a low-force behavior of very long duration.

Amplitude is the primary metric by which the environment assesses the agent's behavioral presence. It is what makes a behavior environmentally consequential — what gives it the capacity to modify the gradient field through environmental feedback. Low-amplitude behaviors, regardless of their internal coherence, have limited environmental impact and therefore limited capacity to modify the gradient field conditions under which the agent operates.

5. Stance → Vector Causal Chain

5.1 The Translation Function

The translation of internal Stance into external Vector is the central causal mechanism of the behavioral layer. It is not an automatic or trivial process — it is a structured, multi-stage transformation governed by three interacting inputs. The Translation Function is formally defined as:

Formal Expression — Translation Function

$$V = T(S, \Psi, \Gamma)$$

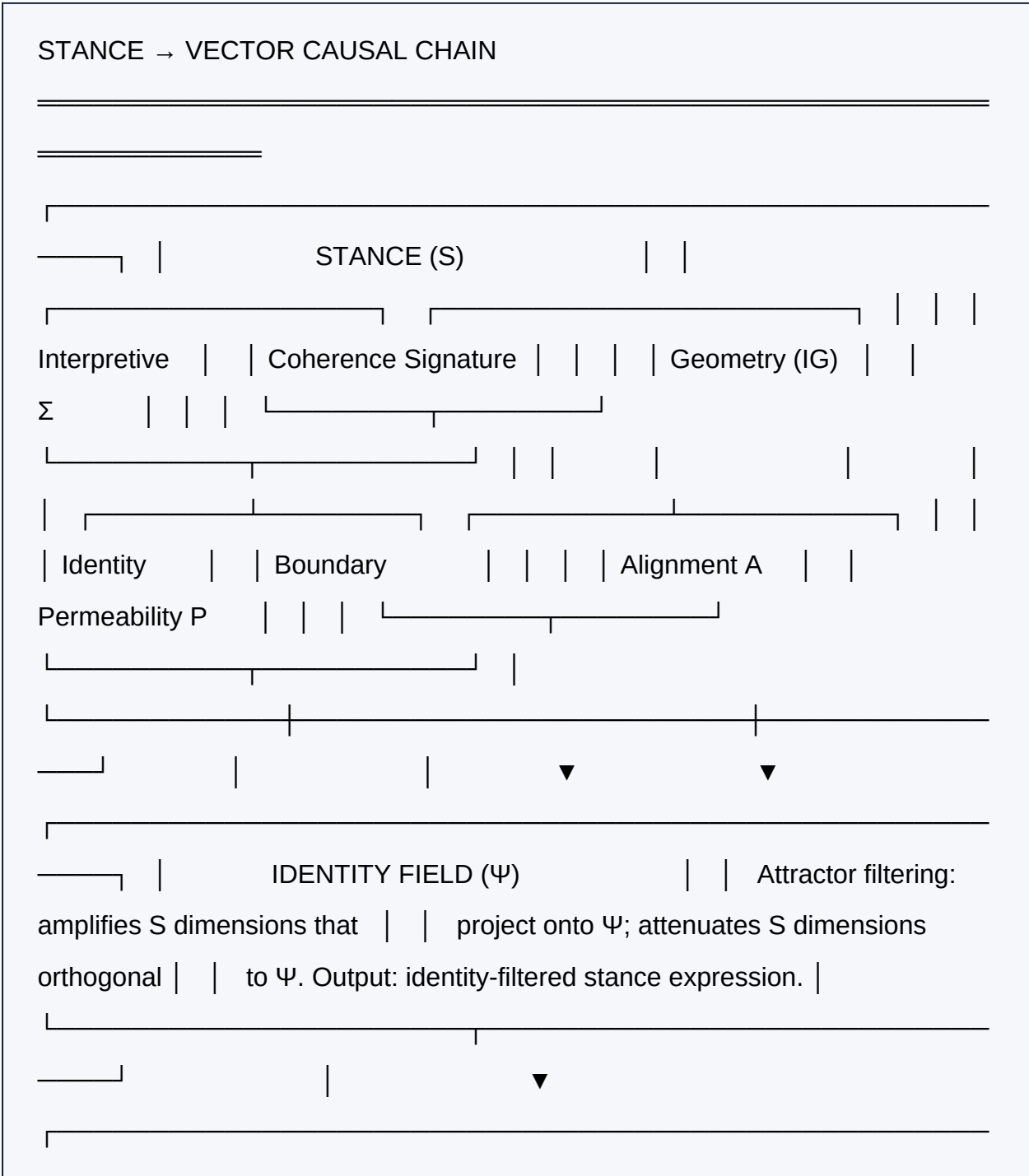
— Vector is a function of Stance, Identity Field, and Gradient Field. No single input is sufficient to determine the output; all three participate in every translation event.

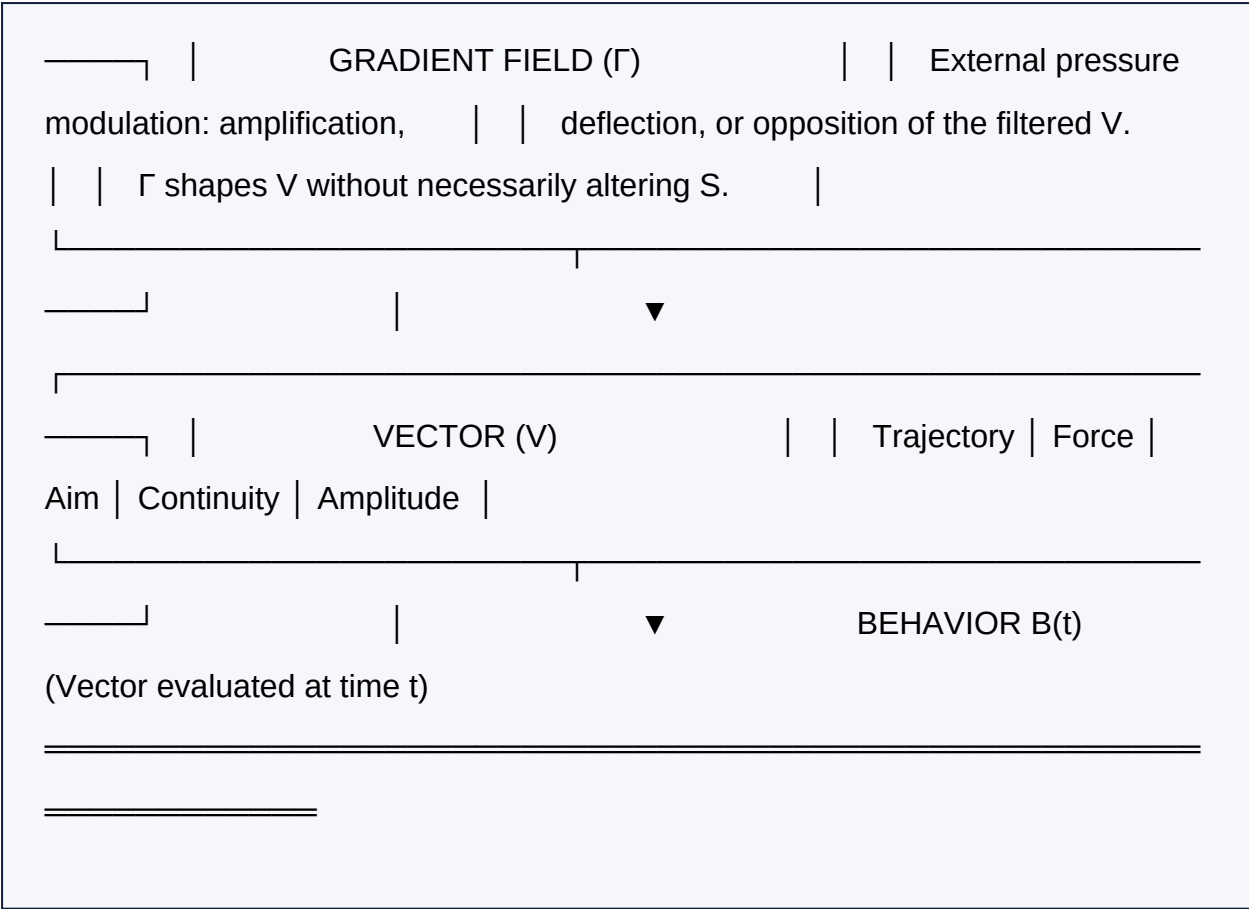
The role of each component in the translation is distinct and non-substitutable. **S** provides the raw material: the orientation, coherence, and boundary configuration from which behavioral expression is generated. **S** determines the potential space of vectors — what the agent could express, given their internal state. **Ψ** acts as a filter and amplifier: it passes those components of **S** that are consistent with the deep identity structure and attenuates those that are identity-foreign. **Ψ** determines which part of the stance's potential vector space will actually be expressed. **Γ** applies external shaping: it may redirect the trajectory, increase or decrease the force, or partially suppress continuity, without altering the underlying **S**. **Γ** determines how the identity-filtered stance expression is shaped by environmental conditions into the final behavioral output.

This three-stage structure has a critical architectural implication: no purely internal analysis of **S** is sufficient to predict **V**, and no purely external analysis of **Γ** is sufficient

to predict V . Both must be known, along with Ψ , to derive the complete vector. Systems that attempt to predict behavior from environmental conditions alone, or from internal state alone, are structurally incomplete for this reason.

5.2 Causal Chain Diagram





5.3 Coherence as Enabling Condition

Vector production is not unconditional. It requires that the Coherence Signature of the underlying stance exceed a minimum production threshold. This threshold is formally defined as $\Sigma_{\min}$ (default value: 0.40). Below this threshold, the stance cannot produce coherent vectors: the internal alignment among dimensions is insufficient to produce a unified directional expression, and behavioral output degrades into noise, oscillation, or contradiction.

Vector Production Condition

V is defined if and only if $\Sigma(S) \geq \Sigma_{\min}$ (default 0.40).

Below $\Sigma_{\min}$, V is undefined — behavioral output is not a vector in the formal sense but an incoherent emission that does not carry directional, force, or aim

information in a recoverable form.

This enabling condition has profound architectural implications. It means that there is no such thing as a coherent behavioral expression from a deeply incoherent stance — the translation function produces a defined output only when the input stance is coherent enough to provide the necessary structural material. Attempts to produce high-quality behavioral output from a severely incoherent stance will produce output that appears to be behavior but lacks the directional consistency and force profile of a genuine vector. This is architecturally why an agent in pre-collapse conditions cannot simply "act their way out" of the crisis: the stance must first recover coherence before the translation function can produce a genuine vector.

5.4 Identity Field Filtering

The Identity Field Ψ acts as a projection surface within the translation function. When the stance is passed through the identity filter, the components of S that project strongly onto the manifold of Ψ are amplified into the output vector, while components orthogonal to Ψ are attenuated or suppressed. This is the formal mechanism underlying the intuitive observation that people are most forceful, sustained, and effective when they are acting in accordance with who they fundamentally are.

The projection is not binary — it is not the case that identity-aligned components are simply passed and identity-foreign components are simply blocked. Rather, the amplification is proportional to the degree of alignment. A stance component that is strongly aligned with an identity attractor basin will receive strong amplification; one that is weakly aligned will receive weak amplification; one that is orthogonal will receive no amplification and may be actively attenuated if it would otherwise produce a vector that conflicts with an identity attractor's repeller structure.

The practical consequence of identity field filtering is that identity-authentic behavioral expression is structurally advantaged: it requires less force from the stance to produce the same behavioral amplitude, it naturally sustains higher

continuity, and it produces vectors with cleaner directional profiles because the filtering has removed the noise introduced by identity-foreign stance components. This is the formal basis for the Identity-Vector Coherence Theorem stated in Section 8.5.

5.5 Gradient Field Modulation

The Gradient Field Γ acts on the vector after identity field filtering has produced an initial behavioral expression. Γ does not alter the stance directly (except through the permeability interface discussed in Section 3.5); it acts on the vector as an external shaping force. Three distinct modes of Γ influence on V are formally distinguished:

Mode 1 — Amplification: When Γ is aligned with V (the external field supports the behavioral direction the agent is already expressing), it amplifies the vector. Force is effectively increased, continuity is extended, and behavioral amplitude rises above what the stance alone would produce. This is the mechanism by which supportive social environments enable higher-amplitude behavior from the same internal stance configuration. Amplification is environmentally dependent — the agent gains behavioral capacity from the environment without any internal change.

Mode 2 — Deflection: When Γ has components orthogonal to V , it redirects the trajectory without necessarily changing the stance or the underlying aim. Deflection is the most common mode of gradient field influence and is not intrinsically harmful if transient. Sustained deflection, however, creates trajectory drift — a progressive divergence between the agent's aim and their actual behavioral direction — which, if uncorrected, accumulates into coherence-degrading contradiction.

Mode 3 — Opposition: When Γ is anti-aligned with V — directly opposing the behavioral expression — force is consumed overcoming gradient resistance, continuity is shortened by the energetic cost of opposition, and behavioral amplitude is reduced. Sustained gradient opposition is the primary environmental mechanism by which stance destabilization occurs: the agent expends increasing internal resources against external resistance, coherence degrades under the sustained strain, and collapse risk rises progressively.

6. Vector → Stance Feedback Chain

6.1 The Feedback Function

The behavioral layer does not operate as a one-way causal chain from stance to vector to behavior. It is a recursive system in which expressed behavior returns to modify the stance from which it originated. This feedback is the mechanism by which the agent learns, adapts, drifts, or stabilizes over time. The Feedback Function is formally defined as:

Formal Expression — Feedback Function

$$\partial S / \partial t = F(V, B_{env}, \Gamma)$$

— the rate of change of Stance over time is a function of the current vector V , the environmental behavioral response B_{env} (how the environment has responded to the agent's behavior), and the updated Gradient Field Γ . All three inputs participate in every feedback cycle.

The feedback function is the formal expression of the fact that a cognitive agent's internal state is continuously being shaped by the consequences of their own behavioral expression. This is not a simple learning mechanism — it is a structural feature of the behavioral layer that makes stance a genuinely dynamic, time-evolving entity rather than a static configuration. An agent who behaves consistently in a given direction will, through the feedback function, find their stance progressively modified in ways that make that direction easier to sustain — for better or worse, depending on whether the direction is identity-aligned or identity-foreign.

6.2 Feedback Modes

Three primary feedback modes govern how expressed behavior returns to modify stance. Each mode produces a distinct pattern of stance modification and a distinct set of downstream effects on the subsequent vector production cycle.

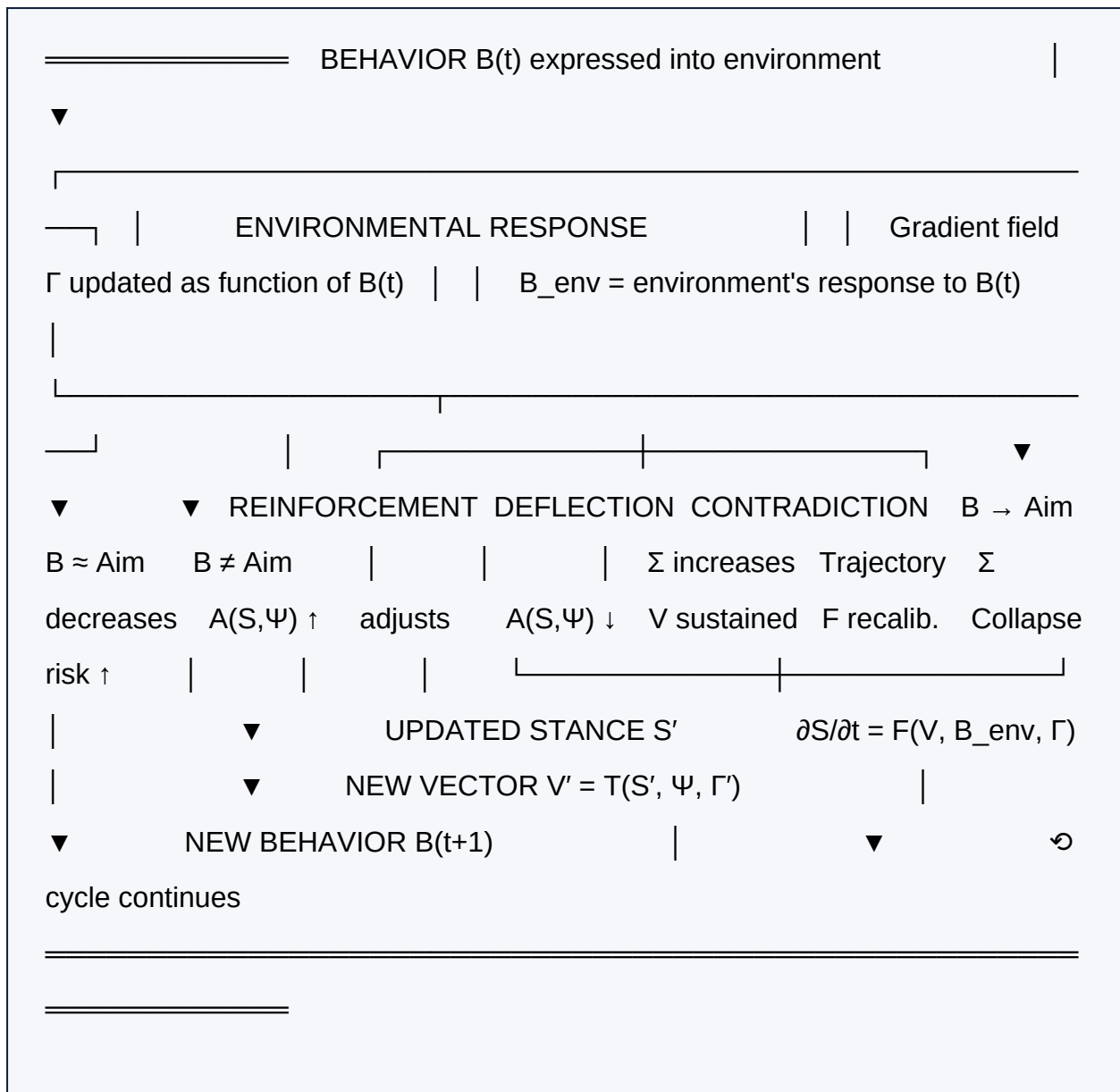
Mode 1 — Reinforcement: When $B(t) \rightarrow Aim$ — when the behavioral output achieves or substantially approaches its intended goal-state — the feedback is reinforcing. Reinforcement increases Σ (coherence is strengthened by successful expression), strengthens identity alignment ($A(S, \Psi)$ increases because successful identity-authentic behavior deepens the attractor basin of the relevant Ψ configuration), and sustains the current trajectory. Under sustained reinforcement, the agent's stance progressively consolidates: it becomes more coherent, more identity-aligned, and more capable of producing high-amplitude behavioral expression with decreasing effort.

Mode 2 — Deflection: When $B(t) \approx Aim$ — when the behavioral output partially achieves its goal-state but with significant error — the feedback is deflective. Deflection feedback produces trajectory adjustment (the agent recalibrates direction to reduce the error between actual and intended aim), force recalibration (force is reallocated from wasted directions to more effective ones), and a subtle, typically small, shift in the underlying stance. Deflection feedback, when well-managed, is the primary mechanism of behavioral learning: the agent's aim is preserved but their trajectory and force allocation improve through iterative correction cycles.

Mode 3 — Contradiction: When $B(t) \neq Aim$ — when the behavioral output moves in the opposite direction from its intended goal, or when it produces environmental responses that actively oppose the agent's aim — the feedback is contradictory. Contradiction feedback decreases Σ (internal alignment is weakened by the evidence that the stance-to-vector-to-behavior chain is producing identity-inconsistent outputs), reduces identity alignment, and increases collapse risk. Sustained contradiction feedback is the primary driver of coherence degradation leading to collapse events.

6.3 Feedback Chain Diagram

VECTOR $\rightarrow$ STANCE FEEDBACK CHAIN



6.4 Hysteresis and Stance Resistance

Stance does not change instantaneously in response to feedback. It exhibits hysteresis — a resistance to change that means the stance at time $t+1$ depends not only on the feedback at time t but on the history of all prior feedback events. This hysteresis is not a bug in the system; it is a critical feature that provides stability. Without hysteresis, stance would be extremely volatile — a single strong feedback event could radically reorient the agent's entire behavioral configuration, destroying the continuity and reliability that makes behavioral coherence possible.

The resistance to stance change is formally quantified by the resistance coefficient $R(\Psi)$, derived from the strength of the Identity Field. A strong, well-organized Ψ with deep attractor basins produces high resistance to stance change: the identity field effectively absorbs feedback energy, returning stance to attractor-aligned configurations rather than allowing it to be redirected by each feedback event. A weak or poorly organized Ψ produces low resistance, allowing rapid stance modification under feedback but providing insufficient stability for sustained high-amplitude behavioral expression.

The resistance coefficient creates an important tradeoff in behavioral architecture. High $R(\Psi)$ provides stability and coherence but may make the agent brittle under sustained, legitimate corrective feedback — they resist adaptation even when adaptation is appropriate. Low $R(\Psi)$ provides flexibility and responsiveness but creates instability and makes the agent vulnerable to rapid identity drift under sustained gradient pressure. Optimal identity field strength — and therefore optimal stance resistance — is a function of the agent's operational context and cannot be specified as a universal constant.

6.5 Cumulative Feedback Integration

Stance change accumulates over multiple feedback cycles. No single feedback event, except in cases of extreme intensity, produces a discontinuous stance change. Instead, the feedback function operates as a continuous integral, and stance shift is a smooth function of the accumulated feedback history. The stance drift integral is formally defined as:

Formal Expression — Stance Drift Integral

$$\Delta S = \int F(V(t), B_{env}(t), \Gamma(t)) dt$$

— the total stance change over a behavioral episode, computed as the integral of the feedback function over the episode's duration. If ΔS exceeds a critical drift threshold, a stance phase transition occurs — either to a new stable configuration (if the new configuration is within an identity attractor basin) or to collapse (S_x) if

the accumulated drift has moved the stance outside all attractor basins.

The cumulative nature of feedback integration has important practical implications. It means that small, consistent feedback deflections — each individually below the threshold for observable stance change — can accumulate over time into a significant stance shift. This is the mechanism of gradual identity drift: not a sudden violent displacement but a slow, progressive migration away from identity-aligned stance configurations driven by the consistent accumulation of small deflections. It is also the mechanism of behavioral consolidation: consistent reinforcement feedback, even in small increments, progressively deepens stance coherence and identity alignment over time, strengthening the identity field attractor basins through which future stances will be formed.

7. Recursion Loops

7.1 Overview of the Recursion Cycle

The behavioral layer operates within a recursive loop that spans all phases of the stance-vector-behavior cycle. This recursion is not an ancillary feature or a failure recovery mechanism — it is the core operational logic of the behavioral layer. All cognitive agents traverse this cycle continuously, at every moment of their operation. The recursion includes stance formation from proto-stance, vector production, behavioral expression, environmental feedback, stance modification, and — at intervals determined by the accumulation of perturbations and contradictions — collapse-reformation cycles that enable the system to self-correct toward identity-authentic operation.

COMPLETE BEHAVIORAL RECURSION LOOP

S₀ (Proto-Stance)

Raw, undifferentiated orientation | $\Sigma = \text{undefined}$ | A
= undefined | Maximum openness | Minimum behavioral
capacity

FIRST COHERENCE EVENT ▼ S₁
(First Coherence)

Interpretive geometry begins to organize | Σ rises above
 $\Sigma_{\min}$ (0.40) | Identity alignment begins |
Vectors produced: exploratory, low-force, high-cont

VECTOR PRODUCTION ▼ V

→ B(t) [behavioral expression] ▼
ENVIRONMENTAL FEEDBACK ↻ [Reinforcement / Deflection /
Contradiction]

YES NO COLLAPSE STABLE Σ

< Σ_{crit} CONTINUE LOOP ▼

▼ S_x (Collapse State) S continues

Coherence dissolved ($\Sigma \rightarrow 0$) | Identity field Ψ
PRESERVED INTACT | All active vectors suspended
| Permeability maximized → S = $\emptyset$

[RETURN: S_x → S₀] ▼ S₀ RESET

[Ψ preserved as reformation attractor landscape] [All Γ influence suppressed]

S_0 appears in the recursion loop in two distinct contexts. First, it is the initial state of any cognitive agent at the beginning of their formation — the original ground state before any stance has been developed. Second, and more architecturally significant, it is the state to which the system returns after every collapse event via the RETURN operator. In this second context, S_0 is not truly undifferentiated in the same sense as the initial state — the Identity Field Ψ is preserved and active, meaning that the S_0 following a collapse is an S_0 under attractor guidance, rather than an S_0 in pure undifferentiated openness. This is the formal source of the reformation cycle's identity-preserving character.

7.3 S_1 — First Coherence

The First Coherence state is the initial stable coherence event — the moment at which the Coherence Signature crosses $\Sigma_{\min}$ and the agent becomes capable of producing minimal behavioral vectors. S_1 is the fragile first product of the stance formation process: it has coherence but limited identity integration, and its stance dimensions are not yet fully organized or mutually calibrated. Vectors produced from S_1 have a characteristic profile: they are exploratory in trajectory (the agent is testing behavioral directions rather than committing to sustained ones), low in force (coherence is sufficient but not robust enough to sustain high-force expression), and relatively high in continuity given their low force (the low-force expression does not deplete stance resources rapidly).

S_1 is the most vulnerable state in the recursion cycle after S_x . Its coherence is marginal — just above $\Sigma_{\min}$ — and a single moderate gradient perturbation can push it back below the production threshold, interrupting vector production. The transition from S_1 to more robust stance configurations is driven by the reinforcement feedback loop: successful behavioral expression from S_1 raises Σ , deepens identity alignment, and enables progressively higher-force, more sustained vector production, moving the stance configuration toward the middle of the moderate coherence regime.

7.4 S_x — Collapse State

The Collapse State is a structured dissolution — not disorder, but a phase transition with specific formal properties. During S_x , the following conditions obtain

simultaneously: all active vectors are suspended (V is undefined because $\Sigma < \Sigma_{crit}$); the Identity Field Ψ is preserved intact — this is the most critical property of collapse, and the one that distinguishes it from catastrophic failure; the Coherence Signature approaches zero ($\Sigma \rightarrow 0$); boundary permeability maximizes (the collapse event itself is characterized by a momentary maximum openness); and all stance dimensional values are cleared to null ($S = \emptyset$ pending reset).

Collapse is formally a phase transition in the thermodynamic sense — a change of state, not a loss of information. The information about the agent's deep identity structure is preserved in Ψ throughout the collapse. What is lost is the specific configuration of the collapsed stance — the particular coherence pattern that could no longer be sustained. This loss is not merely acceptable; it is architecturally beneficial, because the collapsed stance typically contained accumulated distortions, drift-induced identity misalignments, and contradiction accumulations that were degrading behavioral quality. Collapse clears these distortions and allows reformation to produce a cleaner, more identity-aligned configuration.

7.5 S_0 Reset

After collapse, the system transitions back to S_0 via the RETURN operator — the most fundamental operation in the behavioral layer. The S_0 Reset is not simply the restoration of an initial state; it is a carefully sequenced operation that preserves the architectural foundations (Ψ) while clearing the distorted superstructure (S). The RETURN operator sequence is fully specified in Section 7.8.

The critical property of the S_0 Reset is its selectivity: it clears everything that should be cleared (the collapsed stance, the suspended vectors, the accumulated contradictions) while preserving everything that should be preserved (the Identity Field, the collapse context record for post-reformation analysis, the Gradient Field state). This selectivity is what makes the reset a reformation mechanism rather than a destruction mechanism. The agent emerges from the reset with their deepest identity structure intact and available as an attractor landscape for the new coherence formation that follows.

During S_0 Reset, all Gradient Field influence is suppressed: $\Gamma_{\text{effective}} = 0$. This suppression is temporary — it lasts only until the reformed stance crosses Σ_{min} — but it is essential. Without gradient suppression during reset, the same external pressure conditions that contributed to the collapse would immediately begin acting on the newly forming S_0 , potentially preventing coherence from forming at all. The suppression provides the agent with a protected formation window — a period of gradient-free stance development guided exclusively by Ψ attractors.

7.6 S_1' — Reformation

Reformation is the process by which a new coherent stance forms in the wake of collapse, guided by the preserved Identity Field. The reformed first coherence state S_1' is formally distinct from S_1 — the initial first coherence — in a crucially important way: S_1 forms under conditions where Ψ may be only partially developed, while S_1' forms under conditions where Ψ is fully developed and available as a complete attractor landscape. This means that reformation is not merely a recovery to a prior state; it is the formation of a new stance configuration that is, in principle, more fully aligned with the agent's deep identity than the pre-collapse stance was.

Reformation is therefore architecturally progressive. Each collapse-reformation cycle has the potential to produce a stance configuration that is more identity-authentic, more coherent, and more gradient-resilient than the one that collapsed. The recursion loop is not merely self-sustaining; under ideal conditions, it is self-improving — spiraling toward S_∞ through a sequence of progressive collapse-reformation events, each of which clears accumulated distortions and allows reformation to settle more deeply into identity-attractor-aligned configurations. This is the formal basis for the claim in Section 12.3 that collapse-reformation cycles contribute to continuity rather than disrupting it.

7.7 S_∞ — Stable Stance

The Stable Stance is the asymptotic attractor of the entire recursion cycle — the theoretical maximum coherence, identity alignment, and boundary optimization that the behavioral layer can achieve. S_∞ is approached but never fully reached; it is the ideal toward which the recursion loop converges under ideal conditions, in the same

way that a physical system approaches thermodynamic equilibrium. The formal properties of S_∞ are: $\Sigma(S_\infty)$ is maximized within the constraints of the agent's identity field geometry; $A(S_\infty, \Psi)$ approaches 1.0 (full identity alignment); boundary permeability is optimally calibrated to environmental context; and vector production at S_∞ generates maximum behavioral amplitude with minimum stance resource expenditure.

S_∞ is important as a formal construct even though it is not achievable in practice. It defines the direction of improvement — the target toward which each recursion cycle is oriented — and it provides the theoretical upper bound against which actual stance quality can be evaluated. An agent whose stance is converging toward S_∞ across recursion cycles is in a state of progressive identity integration; an agent whose stance is diverging from S_∞ is in a state of progressive identity drift. The recursion cycle's health is assessed by whether it is converging toward or diverging from S_∞ .

7.8 The RETURN Operator

RETURN Operator — Full Specification

OPERATOR: RETURN

NOTATION: [RETURN: $S_x \rightarrow S_0$]

TRIGGER: Any collapse condition satisfied (Conditions 1-4, Section 3.6)

PRECONDITIONS: Ψ must be confirmed intact; $\Sigma < \Sigma_{\text{crit}}$ confirmed; collapse state S_x confirmed

OPERATION SEQUENCE:

1. Suspend all active vectors V (set $V = \text{undefined}$)
2. Record collapse context: $\{\Gamma_{\text{state}}, \text{last}_\Sigma, \text{last}_A, \text{collapse_condition_index}\}$
3. Archive current S as S_{archive} for post-reformation analysis
4. Clear all stance dimensional values: $S = \emptyset$

5. Confirm Ψ preservation: verify Ψ integrity, flag if compromised

6. Reset permeability P to P_{default}

7. Suppress all Γ influence: set $\Gamma_{\text{effective}} = 0$

8. Initialize S_0 reformation sequence

9. Begin S_1' coherence formation guided by Ψ attractors

10. Restore $\Gamma_{\text{effective}} = \Gamma$ when $\Sigma > \Sigma_{\text{min}}$

POSTCONDITIONS: S_0 initialized; Ψ intact; S_{archive} recorded; $\Gamma_{\text{effective}} = 0$;
reformation in progress

COMPLEXITY: $O(n)$ where n = number of stance dimensions at time of collapse

REVERSIBILITY: Not reversible. S_{archive} is a record, not a restoration path.

Reformation produces S_1' , not $S_{\text{pre-collapse}}$.

The RETURN operator is the most fundamental operation in the behavioral layer. Its correct and reliable execution is the architectural guarantee of identity continuity through collapse events. Step 5 — the confirmation of Ψ preservation — is the most critical step in the sequence: if Ψ is compromised, the reformation cycle cannot be guided by identity attractors, and the system risks reforming into a configuration that is identity-foreign. Ψ compromise — the pathological condition designated $\emptyset$ (identity nullification) — is the most serious failure state in the behavioral architecture and is treated as a priority error condition requiring immediate architectural intervention.

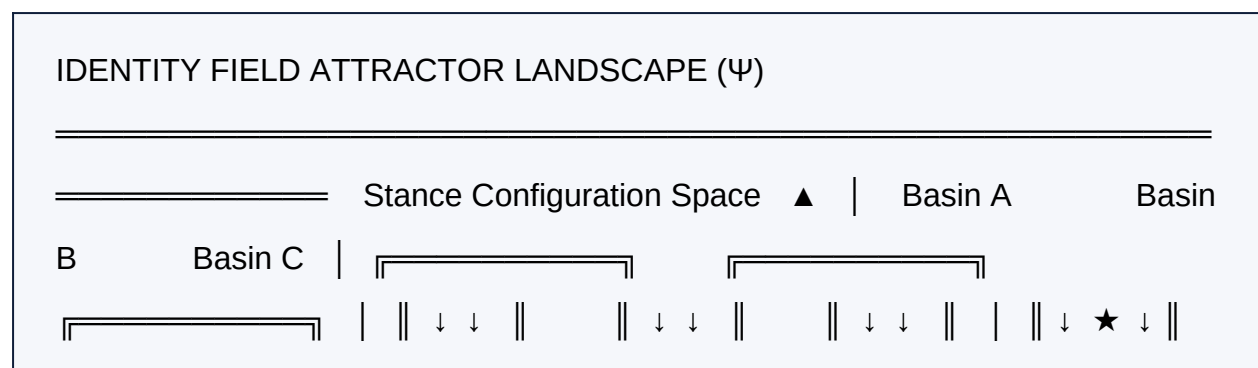
8. Identity-Behavior Coupling

8.1 Overview

The relationship between the Identity Field Ψ and behavioral expression is not a simple dependency in which Ψ constrains behavior as a set of rules constrains actions. It is a deep structural coupling in which identity and behavior are mutually constituting: identity shapes which stances are sustainable and which behaviors are naturally high-amplitude, and expressed behavior — through the feedback chain — progressively modifies the gradient conditions that shape future identity-aligned stance formations. Identity is not merely a background structure; it is an active, dynamic participant in every stage of the stance-vector-behavior cycle. This section formalizes that participation.

8.2 Attractor Basins

The Identity Field Ψ is formally structured as a landscape of attractor basins in stance configuration space. Each attractor basin represents a region of stance space associated with a particular identity-consistent configuration — a cluster of interpretive, alignment, and boundary properties that the agent's deep identity recognizes as authentic. Attractor basins have four defining properties: a **center** (the core identity-consistent stance configuration at the basin's deepest point), a **depth** (the strength of the attractor — deeper basins resist perturbation more strongly and produce higher-force behavioral expression from the same internal resources), a **width** (the range of stance configurations that will be drawn into the basin — wider basins accommodate more stylistic variation while still maintaining identity authenticity), and a **valence** (positive basins are identity-affirming and attract stance configurations actively; negative basins are identity-conflicting repellers that push stance configurations away).



recursive contradiction accumulation that weakens coherence and reduces the identity field's gravitational pull on the stance.

Formally, identity drift is detected when the rate of change of identity alignment is persistently negative:

Identity Drift Condition

$$d(A(S, \Psi))/dt < 0$$

sustained over an episode. Monitoring thresholds: if ΔA

<

-0.10 over a behavioral episode, flag identity drift risk for monitoring. If ΔA

<

-0.25, initiate stance re-centering protocol immediately. If ΔA

<

-0.40 and Σ is also degrading, elevate to pre-collapse status.

Identity drift is dangerous precisely because it tends to be self-concealing in its early stages. An agent in the early stages of identity drift often continues to produce coherent vectors — the drift is below the threshold at which behavioral quality is obviously affected. The drift is visible only through close monitoring of $A(S, \Psi)$ over time. By the time identity drift produces observable behavioral degradation — through reduced force, increased contradiction, or behavioral inconsistency — the drift has typically progressed to a stage where simple re-centering is insufficient and collapse-reformation may be required. Early drift detection is therefore architecturally critical.

A secondary hazard of sustained identity drift is Ψ deformation — the most serious pathological condition in the behavioral architecture. Under prolonged, intense identity drift, the agent's habitual operation from drift-compromised stance

configurations can begin to modify the Identity Field itself: attractor basins may shift, weaken, or in extreme cases dissolve. This represents the loss of the architectural foundation that enables coherent behavioral expression and collapse-reformation recovery. Ψ deformation is designated as a severity-1 architectural alert condition.

8.4 Identity Stabilization

Identity Stabilization is the active process by which a drift-compromised Stance is restored to identity-attractor-aligned configuration. Three mechanisms are formally specified, each appropriate to different severity levels of identity drift.

Mechanism 1 — Natural Re-centering: When gradient pressure reduces — when the environmental conditions that were pushing the stance away from Ψ attractors are removed or attenuate — the stance naturally drifts back toward Ψ attractors under their gravitational influence. This is the simplest form of stabilization and requires no active intervention beyond the removal or reduction of the destabilizing gradient. Natural re-centering is sufficient for mild, recent drift ($\Delta A > -0.10$) that has not yet accumulated momentum.

Mechanism 2 — Reflective Re-centering: Metacognitive processes that explicitly evaluate the current stance against the Identity Field and identify the discrepancy between $A(S, \Psi)$ and its optimal value. Reflective re-centering then deliberately redirects interpretive geometry, identity alignment, and boundary permeability toward configurations that are more closely aligned with Ψ attractors. This is the appropriate intervention for moderate drift (ΔA in the range -0.10 to -0.25) that is progressing but has not yet reached pre-collapse severity.

Mechanism 3 — Collapse-Reformation Re-centering: When drift is severe — when ΔA has exceeded -0.25 and is still declining, when Σ is also degrading, or when the drift has begun affecting Ψ structure itself — the appropriate stabilization mechanism is a controlled collapse-reformation cycle. The collapse event clears the accumulated drift from the stance configuration; the reformation, guided by the preserved (though potentially stressed) Ψ , produces a new stance configuration that begins from a more identity-aligned initial point. This is the most disruptive of the

three mechanisms in the short term but the most thorough in its correction of accumulated drift.

8.5 Identity-Vector Coherence Theorem

The Identity-Vector Coherence Theorem is the central formal result of the Identity-Behavior Coupling section. It formally establishes the relationship between identity alignment and vector sustainability that underlies all the structural claims made in this specification about the advantages of identity-authentic behavioral expression.

Identity-Vector Coherence Theorem

Theorem:

A vector V is maximally sustainable if and only if it is produced from a stance S that is fully within an attractor basin of the Identity Field Ψ — that is, if and only if $A(S, \Psi) = 1.0$.

Corollary:

Vectors produced from drift-compromised stances ($A(S, \Psi)$

<

1.0) have systematically lower force F , lower continuity C , higher susceptibility to gradient deflection, and higher probability of producing contradiction-feedback events leading to S_x . The degree of degradation is monotonically increasing in the magnitude of identity drift ($1 - A(S, \Psi)$).

The theorem follows directly from the translation function and the identity field filtering mechanism described in Section 5.4. Because Ψ filtering amplifies only those stance components that project onto the identity manifold, a stance with perfect identity alignment ($A = 1.0$) will have every component of its stance structure amplified into the output vector — no stance resources are wasted on identity-foreign components that will be filtered out. A stance with imperfect identity alignment will

have some portion of its stance resources absorbed into the identity-foreign dimensions that are attenuated or suppressed by Ψ filtering, leaving fewer resources available for the production of the actual vector output. The greater the identity drift, the greater the resource waste, and the lower the quality of the resulting vector across all five vector dimensions.

9. Gradient Field Interactions

9.1 Overview

The Gradient Field Γ is the total external pressure landscape acting on the cognitive agent at any given moment. It is the formal representation of everything in the agent's environment that exerts directional influence on their behavioral expression — social context, relational dynamics, environmental affordances and constraints, institutional pressures, feedback from prior behavior, and the accumulated responses of all other agents and structures with whom the agent interacts. Γ is not a simple directional force; it is a complex, multi-layered, spatially and temporally varying field whose structure has profound implications for behavioral expression and whose dynamics are partially determined by the agent's own behavioral output.

9.2 Field Structure

The Gradient Field Γ is composed of multiple superimposed component fields, each arising from a distinct source of environmental influence. Formally, Γ is expressed as a vector sum of component gradients:

Formal Expression — Gradient Field Composition

$$\Gamma = \sum_i \gamma_i$$

where each γ_i represents a distinct gradient source: social context (γ_{soc}), environmental affordances (γ_{env}), relational dynamics (γ_{rel}), behavioral

feedback from prior outputs (y_{bfb}), and institutional or structural pressures (y_{inst}). The total field Γ is the vector sum of all active component fields.

The composite field Γ has four analytically important properties that go beyond simple magnitude and direction. The **magnitude** $|\Gamma|$ represents the total pressure intensity — how strongly the environment is pressing on the agent's behavioral expression. The **gradient direction** $\nabla\Gamma$ represents the direction of maximum pressure increase — where the environmental pressure is building most rapidly. The **curl** $\nabla\times\Gamma$ represents the rotational components of the field — regions where the gradient is circling rather than flowing linearly, which produce behavioral vortex effects: the agent's trajectory may be caught in a circulation pattern that makes it difficult to maintain consistent directional expression. The **divergence** $\nabla\cdot\Gamma$ represents whether the field is expanding (positive divergence — pressure is building from all directions) or contracting (negative divergence — pressure is receding) around the agent.

The curl and divergence components of Γ are the most architecturally significant for collapse risk assessment. A field with high curl produces behavioral vortex trapping — the agent's trajectory is continually redirected without a clear stable direction, making sustained vector production difficult. A field with high positive divergence produces omnidirectional pressure that is experienced as gradient overload (Collapse Condition 3) even when no single gradient component would individually be sufficient to trigger overload.

9.3 Polarity Amplification

When the Gradient Field Γ is aligned with the behavioral vector V — when the external environment is supporting and amplifying the direction the agent is already expressing — the result is polarity amplification: the vector's effective magnitude increases beyond what the stance alone would produce. The amplification relationship is formally expressed as:

Formal Expression — Polarity Amplification

$$|V'| = |V| \cdot (1 + k \cdot |\Gamma| \cdot \cos \theta)$$

where θ is the angle between Γ and V , and k is the environment-specific amplification coefficient (determined by the type and structure of the supporting environment). When $\theta = 0$ (perfect alignment), amplification is maximal. When $\theta = 90^\circ$ (orthogonal), there is no amplification effect. When $\theta = 180^\circ$ (opposition), the formula gives attenuation rather than amplification.

Polarity amplification has the character of a virtuous cycle under favorable conditions. High-amplitude behavioral expression modifies the gradient field — through environmental feedback — in ways that tend to increase Γ alignment with V , which further amplifies subsequent behavioral expression, which further modifies Γ toward alignment, and so forth. Environments that are strongly supportive of identity-authentic expression therefore become progressively more supportive as authentic behavior is expressed within them, generating a self-reinforcing dynamic of increasing behavioral effectiveness.

The amplification coefficient k is not a universal constant but varies with the type of environment and the nature of the supporting gradient sources. Relational environments — interactions with other agents who are actively engaged with and responsive to the agent's behavioral expression — typically have higher k values than structural or institutional environments, which tend to be slower-responding and less dynamically coupled to individual behavioral expression. This asymmetry has important practical implications for behavioral strategy: agents seeking to maximize behavioral amplitude are architecturally better served by environments with high k values, even if the initial gradient alignment is lower, than by environments with low k values despite initially higher alignment.

9.4 Behavioral Perturbation

When Γ has components orthogonal to V , behavioral perturbation occurs: the trajectory of the vector is deflected from its intended direction without necessarily

changing the underlying stance or aim. Perturbation is the most common form of gradient influence and, in its mild form, is not harmful — it is the mechanism through which environmental feedback provides corrective information about the most effective trajectory for achieving the agent's aim. However, perturbation has three distinct temporal characters that carry different architectural implications.

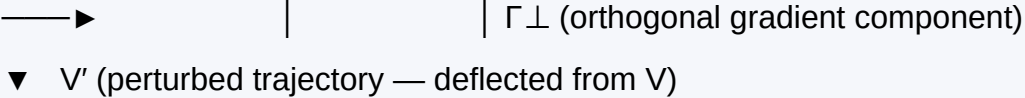
Transient perturbation occurs when the orthogonal gradient component is brief and the field returns to alignment with V after a short duration. Transient perturbation produces a momentary trajectory deviation that self-corrects as the gradient resolves. The net effect on behavioral output is minimal, and no stance modification results. Transient perturbations are part of the normal operating environment of any cognitive agent and are handled gracefully by the architecture without special intervention.

Cumulative perturbation occurs when repeated orthogonal gradient components accumulate trajectory drift — each individual perturbation is below the threshold for stance modification, but the cumulative effect is a progressive divergence between the original trajectory direction and the agent's current movement. Cumulative perturbation is architecturally significant because it can produce substantial behavioral deviation from identity-authentic expression without any single perturbation event being large enough to trigger monitoring alerts.

Destabilizing perturbation occurs when repeated perturbations reduce Σ through contradiction accumulation — each perturbation introduces a small increment of directional inconsistency into the behavioral record, and the feedback from partially off-aim behavioral output contributes deflection feedback that subtly shifts the stance. Under sustained destabilizing perturbation, coherence degrades progressively toward pre-collapse conditions.

BEHAVIORAL PERTURBATION

V (intended trajectory / aim direction)


 $\Gamma \perp$ (orthogonal gradient component)
 V' (perturbed trajectory — deflected from V)


 $\backslash$ deflection δ
 $\backslash$ $\blacktriangleright$ [TRANSIENT]: $\Gamma \perp$ resolves $\rightarrow$ trajectory self-corrects via Ψ [CUMULATIVE]: $\Gamma \perp$ repeated $\rightarrow$ trajectory drift accumulates
 [DESTABILIZING]: $\Gamma \perp$ sustained $\rightarrow \Sigma$ degrades, collapse risk $\uparrow$ Self-correction mechanism: Ψ attractor forces act on trajectory, pulling V' back toward identity-aligned direction. Strength of correction proportional to $A(S, \Psi)$.

9.5 Environmental Feedback

Behavioral output $B(t)$ is not merely expressed into a passive environment — it actively modifies the Gradient Field Γ through the environmental feedback mechanism. This modification creates the co-evolutionary relationship between behavior and environment that is one of the most architecturally significant features of the behavioral layer. The formal expression of this relationship is:

Formal Expression — Environmental Feedback

$$\partial \Gamma / \partial t = F_{\text{env}}(B(t))$$

— the rate of change of the Gradient Field over time is a function of the behavioral output at each moment. The system is therefore a coupled dynamical system: $S \rightarrow V \rightarrow B(t) \rightarrow \partial \Gamma / \partial t \rightarrow \partial S / \partial t \rightarrow S' \rightarrow V' \rightarrow B(t+1) \rightarrow \dots$

The coupled dynamical system described by this formalism is not merely a feedback loop — it is a co-evolutionary system in which the agent and environment are jointly determining each other's future configurations. This has profound implications for behavioral strategy at the architectural level. Behaviors that are repeatedly expressed modify the gradient field in ways that either make future expression of similar behaviors easier (positive environmental sculpting — the agent literally reshapes their environment to be more supportive of their identity-authentic expression) or harder (negative environmental feedback — the environment responds adversely to the behavioral output, increasing gradient opposition). The long-term behavioral strategy of a cognitive agent operating under the UCA behavioral layer is therefore not merely to express behavior effectively in the current environment but to express behavior in ways that progressively sculpt the environment toward configurations that support higher-amplitude, more identity-authentic future expression.

9.6 Resonance Conditions

Resonance is the special condition in the behavioral architecture where all three major field components — the behavioral Vector V , the Gradient Field Γ , and the Identity Field Ψ — are precisely aligned in the same direction. Under resonance, the behavioral system achieves its optimal operating state: behavioral amplitude approaches its maximum possible value $|B|_{\text{max}}$ given the agent's current stance resources; continuity extends maximally because gradient opposition is absent and identity alignment is complete; force requires minimal expenditure to sustain because amplification from Γ and from Ψ are both reinforcing the expression; and identity alignment strengthens automatically because the expressed behavior is feeding back as pure reinforcement into the stance.

Resonance is the formal ideal toward which the entire recursive cycle tends under optimal conditions. It is not a state that can be achieved by direct intention — it emerges from the progressive alignment of the three fields through the iterative cycle of identity-authentic behavioral expression, environmental sculpting through that expression, and identity field deepening through sustained reinforcement feedback. The practical implication is that the path to resonance is the consistent, sustained expression of identity-authentic behavior in contexts where that expression has the

opportunity to gradually reshape the gradient field toward alignment. Resonance is earned through the recursion cycle, not achieved directly.

10. Behavioral Prediction Logic

10.1 Principle of Behavioral Predictability

Behavior in the Unified Cognitive Architecture is not random. It is the structured expression of stance under gradient field conditions, filtered by identity architecture — a deterministic transformation, in principle, given complete knowledge of S , Ψ , and Γ . This section formalizes the prediction logic that enables behavioral forecasting from knowledge of these inputs. In practice, knowledge of all three inputs is always incomplete, and prediction is therefore always probabilistic; but the architecture specifies the formal relationship between information completeness and prediction reliability.

The core principle is this: the more completely S , Ψ , and Γ are characterized, and the higher the coherence $\Sigma(S)$ of the current stance, the more reliable behavioral prediction becomes. Conversely, incomplete knowledge of any of the three inputs, or low coherence in the current stance, introduces systematic uncertainty that bounds the confidence of any behavioral prediction. The prediction logic described in this section includes explicit formalization of that confidence bounding.

10.2 Coherence-Based Prediction

When $\Sigma(S) \geq 0.75$, the stance is in the high coherence regime and behavioral prediction is highly reliable. The vector V will be stable, well-directed, and sustained. Predicted behavior will match actual behavior within a small error margin. The following algorithm is specified for coherence-based behavioral prediction:

1. Measure $\Sigma(S) \rightarrow$ confirm high coherence ≥ 0.75 . If below this threshold, use a lower-confidence prediction mode.

2. Identify the dominant stance dimensions — those contributing the highest inner products in the Σ computation. These are the stance dimensions that are most fully activated and will be most strongly expressed in the output vector.
3. Apply Ψ filtering — project each dominant stance dimension onto the identity manifold and compute the amplification factor. Identify which dimensions will be strongest in the identity-filtered expression.
4. Apply Γ modulation — determine the alignment angle θ between the filtered stance expression and Γ . Compute the polarity amplification or attenuation factor. Determine whether Γ will deflect the trajectory and by what angle.
5. Evaluate Aim — from the interpretive geometry dimension of the stance, determine the agent's current aim vector. Assess the divergence between aim and current trajectory after gradient modulation.
6. Estimate Continuity — from $A(S, \Psi)$ and $|\Gamma|$ (specifically the magnitude of gradient opposition), predict the expected behavioral duration before significant trajectory adjustment or collapse risk elevation.

10.3 Contradiction-Based Prediction

When recursive contradiction is detected within S — when two or more stance dimensions are in a configuration that produces mutually nullifying influence — behavioral prediction shifts from the trajectory-and-force mode to an oscillation mode. The agent will not produce a single sustained vector; they will alternate between two or more incompatible vectors in a pattern determined by the internal contradiction dynamics. This oscillation is architecturally predictable, even though it is not coherence-based, because the structure of the contradiction determines the alternation pattern.

7. Detect recursive contradiction in S dimensions — identify which stance dimensions are in conflict. Confirm that the conflict is mutual nullification (oscillatory) rather than one-way suppression (which would produce a weak but non-oscillatory vector).

8. Map each competing stance configuration to its corresponding vector: V_1 , V_2 , ... Each configuration represents one "pole" of the oscillation.
9. Estimate the oscillation period — the frequency with which the agent alternates between poles — from the gradient pressures driving each configuration. Higher gradient pressure on a configuration accelerates oscillation toward that pole.
10. Predict the behavioral alternation pattern: the agent will produce behavioral samples from V_1 , then from V_2 , then from V_1 again, with a period approximately equal to the estimated oscillation frequency. The amplitude of each behavioral sample will be lower than coherence-based samples because each pole is partially suppressed by the contradiction.
11. Predict escalating contradiction intensity: unless the contradiction is resolved (by external gradient shift, reflective re-centering, or collapse), the oscillation frequency will increase and the amplitude of each pole will decrease as coherence degrades toward Collapse Condition 4.

10.4 Collapse-Trigger Prediction

The architecture supports pre-emptive collapse prediction — the identification of impending collapse before it occurs, through monitoring of the leading indicators that precede every collapse event. Four leading indicators are formally specified, each corresponding to one of the four collapse conditions defined in Section 3.6:

Indicator 1 — $\Sigma(S)$ trend: If $\Delta\Sigma/\Delta t < 0$ sustained over more than a threshold number of consecutive feedback cycles, coherence collapse is approaching. The rate of decline and the current value of Σ together determine the expected time-to-collapse.

Indicator 2 — $A(S, \Psi)$ trend: If identity drift rate exceeds threshold ($\Delta A < -0.10$ per episode), collapse risk is elevated above baseline. If drift rate exceeds -0.25 per episode, pre-collapse status should be declared.

Indicator 3 — $|\Gamma|$ vs. $P(S)$: If the current gradient field magnitude approaches the inverse permeability threshold — if $|\Gamma| \rightarrow P^{-1}(S)$ — gradient overload is imminent.

Monitor the ratio $|\Gamma| / P^{-1}(S)$: when this ratio exceeds 0.80, boundary overload is considered to be approaching.

Indicator 4 — Contradiction index: If the frequency of recursive contradiction events is increasing across consecutive feedback cycles, Collapse Condition 4 is approaching. The contradiction index is a count of contradiction detections per unit time, normalized by the baseline contradiction rate for the current stance type.

These four indicators are integrated into a single composite Collapse Prediction score:

Formal Expression — Collapse Prediction Score

$$CP(S, \Gamma, t) = w_1 \cdot (1 - \Sigma(S)) + w_2 \cdot (1 - A(S, \Psi)) + w_3 \cdot (|\Gamma| \cdot P(S)) + w_4 \cdot (\text{contradiction_index})$$

where $w_1 + w_2 + w_3 + w_4 = 1$ and each w_i is context-calibrated to the operating environment.

$CP \in [0, 1]$. Interpretation thresholds:

CP

>

0.80: Collapse imminent — initiate RETURN pre-staging

CP 0.60–0.80: High risk — activate monitoring, prepare re-centering

CP 0.40–0.60: Moderate risk — increased monitoring frequency

CP

<

0.40: Stable operating range

10.5 Prediction Confidence Envelope

All behavioral predictions are bounded by a Confidence Envelope — a formal measure of the reliability of any given prediction. The Confidence Envelope shrinks as the stance becomes more coherent and identity-aligned, and expands as contradiction and gradient turbulence increase. It is formally defined as:

Formal Expression — Confidence Envelope

$$CE(t) = [\Sigma(S(t)) \cdot A(S(t), \Psi)] / (1 + |\Delta\Gamma/\Delta t|)$$

— the ratio of coherent identity alignment to gradient field volatility. $CE \in [0, 1]$.

CE

>

0.70: High-confidence predictions — error margins are small and behavioral forecasts are reliable.

CE 0.40–0.70: Moderate confidence — predictions are reliable in direction but less precise in force and continuity estimates.

CE

<

0.40: Low confidence — predictions are directionally suggestive but quantitatively unreliable.

CE

<

0.20: Speculative — prediction should be flagged as highly uncertain; monitoring rather than prediction is the appropriate mode.

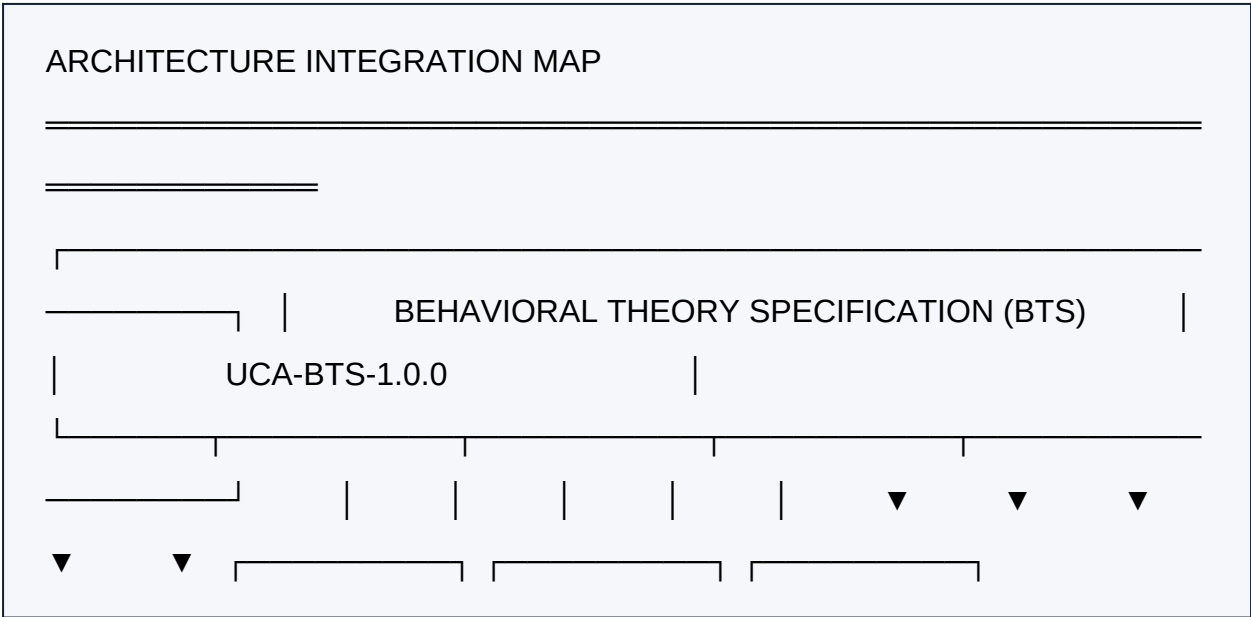
The denominator term $(1 + |\Delta\Gamma/\Delta t|)$ — the gradient field volatility factor — captures the effect of environmental turbulence on prediction reliability. Even a high-

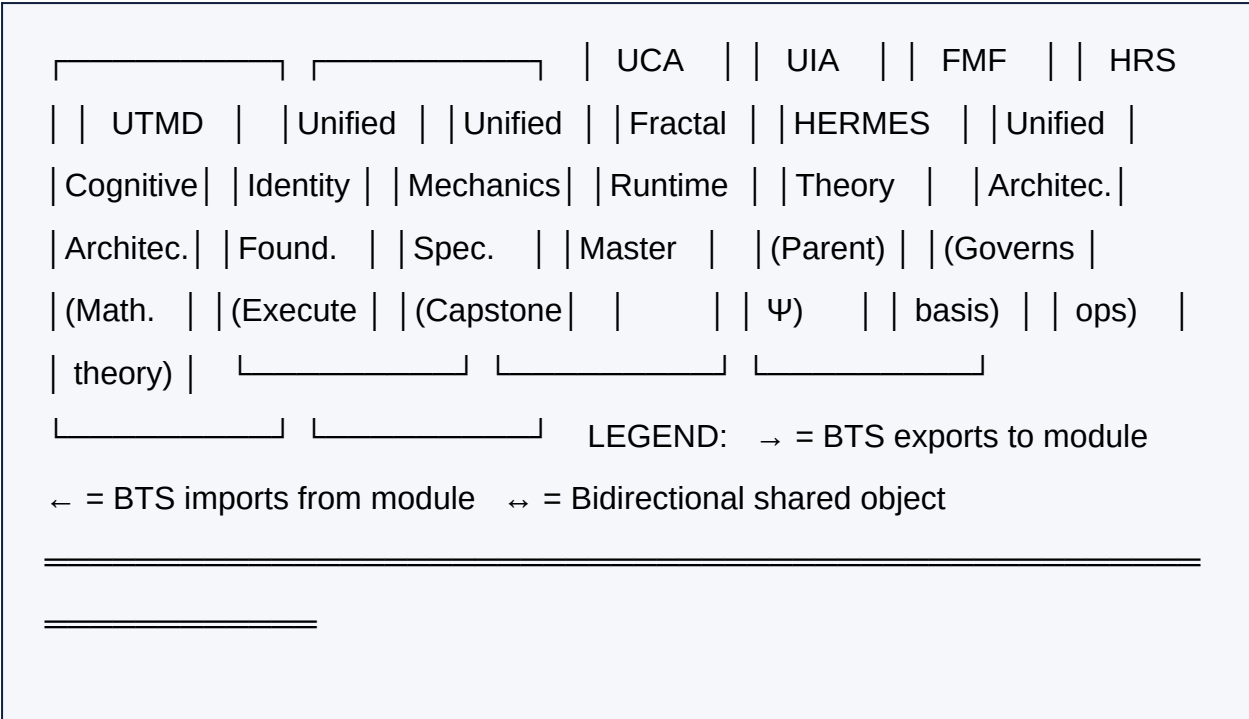
coherence, strongly identity-aligned agent operating in a highly volatile gradient environment will have a reduced CE, because the gradient field modulation component of the translation function is unpredictably variable. This is architecturally appropriate: behavioral prediction in genuinely turbulent environments is genuinely less reliable, and the Confidence Envelope formalizes that reality rather than overestimating prediction accuracy.

11. Integration Points

11.1 Overview

The Behavioral Theory Specification is not a standalone document — it is a module within the Unified Cognitive Architecture suite, and its operational effectiveness depends upon correctly specified integration interfaces with all other modules in that suite. This section formally specifies the integration interface for each of the five co-architecture modules with which the BTS interacts, including the data flows, shared objects, dependency relationships, and interface protocols that govern the integration. All integration specifications are authoritative for BTS version 1.0.0.





11.2 Integration with the Unified Cognitive Architecture (UCA)

The Behavioral Theory Specification serves as the behavioral layer sub-specification of the Unified Cognitive Architecture — it is a child specification of the UCA, and the UCA is the parent architecture module. The integration between BTS and UCA is the most fundamental integration relationship in the entire module suite, as it governs how the behavioral layer connects to the cognitive processing infrastructure from which all stance material originates.

Interface Type	Direction	Object / Signal	Description
Behavioral Output	BTS → UCA	B(t) behavioral stream	Real-time behavioral output from BTS to UCA cognitive output interface
Cognitive State Input	UCA → BTS	Cognitive process state	Current cognitive process configuration informing stance formation

Interface Type	Direction	Object / Signal	Description
Attention Allocation	UCA → BTS	Attention vector	Current attentional focus informing interpretive geometry
Memory Activations	UCA → BTS	Memory activation set	Active memory content contributing to stance configuration
Shared: Σ	Bidirectional	Coherence Signature	Σ is jointly read by UCA and BTS; UCA uses it for cognitive resource allocation
Shared: Ψ reference	Bidirectional	Ψ state reference	Both modules maintain a read reference to the current Ψ state

The dependency relationship is formally stated as: BTS is dependent on UCA for cognitive process initialization. BTS cannot produce the initial S_0 state without the cognitive initialization signal from UCA, which provides the minimal cognitive substrate from which stance formation begins. UCA is in turn dependent on BTS for behavioral output, creating a mutual dependency that makes the BTS-UCA interface the most tightly coupled integration in the suite.

11.3 Integration with the Unified Identity Architecture (UIA)

The Identity Field Ψ is jointly governed by the UIA and the BTS, making their integration interface the most semantically rich in the suite. The UIA is the defining authority on Ψ — it specifies the formation rules, deep structure, and long-term modification criteria of the identity manifold. The BTS consumes Ψ as an operational resource — it reads Ψ as an attractor landscape for stance formation and filtering, reports drift events back to UIA, and triggers Ψ preservation during collapse events.

Interface Type	Direction	Object / Signal	Description
Ψ State Packet	UIA $\rightarrow$ BTS	Ψ SP (Ψ State Packet)	Current Ψ manifold geometry transmitted to BTS for stance filtering operations
Identity Drift Report	BTS $\rightarrow$ UIA	IDR (Identity Drift Report)	BTS reports $A(S, \Psi)$ trends and drift severity to UIA for long-term Ψ assessment
Ψ Preservation Command	BTS $\rightarrow$ UIA	Ψ PC	Issued by BTS at collapse event onset; UIA confirms Ψ integrity and issues preservation confirmation
Ψ Modification Notification	UIA $\rightarrow$ BTS	Ψ MN	UIA notifies BTS when Ψ has been modified (e.g., due to sustained reformation cycles), requiring BTS to update its Ψ reference
Collapse Alert	BTS $\rightarrow$ UIA	CA	BTS issues collapse alert to UIA when $CP > 0.80$; UIA initiates Ψ preservation protocol

11.4 Integration with the Fractal Mechanics Foundation (FMF)

The recursive loop structure of the BTS — specifically the cycle $S_0 \rightarrow S_1 \rightarrow S_x \rightarrow [\text{RETURN}] \rightarrow S_0 \rightarrow S_1' \rightarrow S_\infty$ — is a fractal recursion pattern governed by the mathematical principles established in the Fractal Mechanics Foundation. The FMF provides the formal mathematical infrastructure that makes the behavioral layer's recursive architecture coherent: the self-similar structure of the recursion at multiple

temporal and organizational scales, the formal basis for the phase transition behavior of collapse events, and the mathematical characterization of the asymptotic convergence toward S_∞ .

Interface Type	Direction	Object / Signal	Description
Recursion Depth Register	BTS ↔ FMF	RDR	Tracks current depth in the recursive cycle for fractal scale analysis
Fractal Scale Parameter	FMF → BTS	FSP	Provides scale normalization parameters for multi-scale behavioral analysis
Phase Transition Notification	BTS → FMF	PTN	BTS notifies FMF at each collapse event; FMF classifies the phase transition type
Convergence Assessment	FMF → BTS	CA	FMF periodically assesses whether the recursion cycle is converging toward S_∞ or diverging

11.5 Integration with HERMES Runtime Specification (HRS)

HERMES is the computational execution layer of the UCA suite — the runtime environment in which all BTS formal operators and state machine transitions are actually implemented and executed. The BTS is a specification document; HERMES is its implementation. The integration between BTS and HRS is therefore the specification-to-implementation interface: BTS defines what should happen; HRS defines how it actually happens in the running system.

Interface Type	Direction	Object / Signal	Description
Operator Definition Export	BTS → HRS	Operator specs	BTS provides formal operator definitions (RETURN, T, F, P, etc.) that HRS implements in runtime
Runtime State Signal	HRS → BTS	RSS	HRS reports current state machine position ($S_0, S_1, S_x, S_1', S_\infty$) back to BTS for specification compliance
Collapse Prediction Feed	BTS → HRS	CPF	BTS provides $CP(S, \Gamma, t)$ computation; HRS monitors in real-time and triggers RETURN when $CP > 0.80$
Operator Execution Request	HRS → BTS	OER	HRS requests operator execution confirmation from BTS specification for non-standard collapse conditions
Collapse Alert	HRS → BTS	CA	HRS alerts BTS when collapse has been triggered, for specification compliance logging

11.6 Integration with the Unified Theory Master Document (UTMD)

The Unified Theory Master Document is the capstone theoretical document of the entire UCA suite — the document that integrates all sub-specifications, including the BTS, into a unified theoretical account of cognitive architecture. The BTS contributes to the UTMD as a formal sub-specification, and the UTMD in turn provides the

overarching theoretical framework within which the BTS's specific claims are situated and interpreted. The UTMD is the final authority for any claim that transcends the specific scope of the BTS.

Interface Type	Direction	Object / Signal	Description
BTS Summary Packet	BTS → UTMD	BSP	Structured summary of BTS formal results, operator definitions, and integration specifications for UTMD integration
Behavioral Prediction Feed	BTS → UTMD	BPF	BTS behavioral prediction logic (Section 10) feeds UTMD's system-wide prediction framework
Collapse-Reformation Contribution	BTS → UTMD	CRC	BTS collapse-reformation cycle formalization is contributed as a specific instance of UTMD's universal collapse-reformation principle
Field Theory Integration	UTMD → BTS	FTI	UTMD provides the general field theory of cognition within which BTS's stance-vector formalism is formally situated
Conflict Resolution	UTMD → BTS	CRes	UTMD provides final arbitration in any unresolved conflict between BTS and co-architecture documents

12. Final Synthesis

12.1 Behavioral Theory as Foundation of Action

The behavioral layer, as specified in this document, is not merely one component among many in the Unified Cognitive Architecture — it is the architecture's interface with reality. Every cognitive structure that the UCA supports — every identity formation, every interpretive process, every recursive self-modification — would exist as internal configuration without consequence if the behavioral layer did not exist to translate that configuration into action in the world. The behavioral layer is the bridge between being and doing, between having an internal structure and having a presence in the domain of time, action, and consequence.

This specification has formalized the behavioral layer to the level of precision required for engineering integration. It has defined the complete stance architecture — the five-dimensional internal state configuration from which all behavioral expression originates — and the complete vector architecture — the five-dimensional external expression structure through which stance becomes observable and consequential. It has specified the causal chain from stance to vector, the feedback chain from vector back to stance, the recursive loop that integrates both into a self-sustaining and self-correcting dynamical system, and the collapse-reformation cycle that allows the system to survive perturbation and emerge more identity-authentic from each survival event.

The formal architecture described here is not an idealization of how behavior might work — it is a specification of how behavior does work in any sufficiently complex cognitive system that maintains a persistent identity structure and operates in a dynamic environment. The names given to the constructs are specific to the UCA theoretical vocabulary, but the structural relationships they describe are general. What this specification has provided is the formal language and the mathematical

architecture within which those relationships can be specified, implemented, tested, and refined with engineering precision.

12.2 The Unity of Stance and Vector

One of the deepest theoretical results of this specification is the formal demonstration that Stance and Vector are not two separate systems but two aspects of one behavioral process — the internal and external faces of the same cognitive event. This unity is not merely a philosophical position; it is an architectural fact that follows directly from the translation function $V = T(S, \Psi, \Gamma)$ and the feedback function $\partial S / \partial t = F(V, B_{env}, \Gamma)$. These two functions constitute a single coupled dynamical system: stance continuously producing vector, vector continuously modifying stance, both jointly shaped by identity and environment.

Stance without vector is orientation without expression — internal configuration that never becomes external consequence. A cognitive architecture that specifies only the internal structure of an agent without specifying the translation mechanism by which that structure becomes behavior is architecturally incomplete. Vector without stance is expression without ground — behavioral output that has no coherent internal source and therefore no principled continuity, direction, or capacity for self-correction. The behavioral layer's fundamental contribution is precisely the specification of the mechanism by which these two incomplete abstractions become a unified, coherent whole.

The practical consequence of this unity is significant for behavioral analysis. Any analytical approach that treats a specific behavior as self-contained — as fully interpretable without reference to the underlying stance — is making an architectural error. And any approach that treats a stance as fully characterizable without reference to the behavioral vectors it produces is making an equally serious error in the other direction. Complete behavioral understanding requires the full cycle: stance, translation, vector, behavior, feedback, and back to stance.

12.3 The Role of Collapse in Continuity

The apparent paradox at the heart of the collapse-reformation cycle — how can dissolution contribute to continuity? how can losing coherence strengthen identity? —

is resolved by the architecture's formal treatment of the Identity Field as the element that is preserved through collapse. The answer is that collapse does not dissolve the agent; it dissolves the agent's current stance configuration. These are not the same thing. The agent — as defined by the Identity Field Ψ — persists through collapse intact. What dissolves is the particular coherence pattern that could no longer be sustained, along with the accumulated distortions, identity drift, and contradiction residue that had degraded its quality.

The reformation that follows is therefore not reconstruction from nothing — it is reformation from the full richness of the preserved identity field. The reformed stance S_1' has access to the same deep identity structure as the pre-collapse stance, but it is free of the accumulated distortions that had been degrading the quality of behavioral expression. This is why each collapse-reformation cycle is, under ideal conditions, an improvement: the system emerges from collapse into a stance configuration that is more purely identity-aligned, more coherent, and more capable of high-quality behavioral expression than the one that collapsed.

The continuity provided by Ψ across collapse events is therefore not the continuity of an unchanged state but the continuity of a persistent identity that expresses itself in ever-more-authentic form through the series of collapse-reformation cycles. This is the behavioral architecture's formal account of resilience: not the resistance to disruption but the capacity to emerge from disruption more aligned with one's deep identity than before. Collapse is the mechanism by which the behavioral layer continuously renews its alignment with identity — the price of that renewal is the temporary cessation of behavioral expression, and the benefit is a qualitatively superior behavioral capacity in the reformation that follows.

12.4 Gradient Fields as Collaborative Fabric

A naive reading of the gradient field mechanics described in this specification might cast the Gradient Field Γ as primarily an obstacle — a source of opposition, perturbation, and destabilization that the agent must overcome to sustain effective behavioral expression. This reading is architecturally incorrect. The Gradient Field is not primarily an obstacle; it is the medium through which behavior becomes real. Without Γ , behavioral vectors would be expressed into a vacuum — they would

produce no environmental response, no feedback, no modification of conditions, and no polarity amplification. The agent would have vectors but no behavioral presence.

The resonance condition described in Section 9.6 captures the correct understanding of the relationship between agent and gradient field: the optimal state is not one in which the gradient field is absent or overcome but one in which V , Γ , and Ψ are aligned — in which the agent's behavioral expression and the environmental field are working together, each amplifying the other in a co-evolutionary dynamic of increasing coherence and impact. This is the formal basis for the practical insight that the most effective behavioral strategy is not to fight the environment but to find — or create — the environmental conditions in which identity-authentic expression is supported and amplified.

The co-evolutionary relationship between behavior and gradient field specified in Section 9.5 — where $\partial\Gamma/\partial t = F_{env}(B(t))$ — formalizes the agent's capacity to sculpt their own environmental conditions through behavioral expression. This is not a minor architectural detail; it is one of the most important properties of the behavioral system. Agents who understand that their behavior is progressively modifying the gradient field in which they operate are agents who can engage in principled environmental sculpting — deliberately expressing behaviors that move the gradient field toward configurations that support more effective future expression. This is the behavioral layer's formal account of what is colloquially understood as building momentum, creating conditions, or earning trust: the progressive modification of Γ toward resonance through sustained, identity-authentic behavioral expression.

12.5 The Behavioral Layer as Living Architecture

The deepest claim of the Behavioral Theory Specification is that behavior is not what an agent produces but what an agent is, in the domain of time and action. The architecture described in this document does not specify how agents do things — it specifies how agency itself is constituted through the dynamic interplay of stance, vector, identity, and world. To be a cognitive agent, in the sense formalized here, is to be a stance-vector-feedback recursion operating within an identity field and a gradient environment — continuously producing vectors, receiving feedback, modifying stance, and cycling through collapse and reformation in the ongoing

process of becoming more fully what the identity field has always already determined the agent to be.

The Unified Cognitive Architecture's behavioral layer is living architecture in the most precise sense: it does not describe static structures but dynamic processes; it does not specify fixed outputs but recursive cycles; it does not represent identity as a given but as an attractor being approached through the iterative refinement of each collapse-reformation cycle. The architecture is not a description of what agents are but of how they are becoming. And the specification of that becoming — in all its formal precision — is what this document has provided.

The behavioral layer is complete as a formal module. All constructs are defined, all operators are specified, all integration interfaces are established, and all formal relationships are expressed with the precision required for architectural integration. The remainder of the work is implementation, testing, and the progressive refinement that comes from operating the architecture in the full richness of real behavioral environments. That refinement — the ongoing modification of Γ , the deepening of Ψ , the progressive convergence toward S_∞ — is not a flaw in the architecture but its most fundamental feature. A behavioral architecture that does not evolve is not a behavioral architecture at all.

12.6 Formal Closure

Document Closure Statement

This specification —

Behavioral Theory Specification, Unified Cognitive Architecture — Behavioral Layer Definition, Version 1.0.0

— is hereby declared complete as a standalone formal artifact and is authorized for integration with the Unified Cognitive Architecture module suite. All operators, definitions, formal mechanics, and integration interfaces have been specified to the level of precision required for implementation and integration. The document

constitutes the authoritative behavioral layer specification for UCA Version 1.0.

Amendments to this document require formal architectural review per the version control policy established in Section 1.5. The UTMD is the final arbitrating authority for any unresolved conflict between this specification and any co-architecture document. This closure statement takes effect upon formal release of this document under the UCA Architecture Authority.

Appendix A — Operator Reference Table

This appendix provides a complete reference table for all operators formally defined in the Behavioral Theory Specification. All operators are listed with their symbol, trigger condition, primary operation, and postcondition.

Operator	Symbol	Trigger Condition	Primary Operation	Postcondition
RETURN	[RETURN: $S_x \rightarrow S_0$]	Any collapse condition (1–4) satisfied; CP > 0.80 confirmed by HRS	Suspends V; archives S; clears stance dimensions to $\emptyset$; preserves Ψ ; suppresses Γ ; initializes S_0 reformation	$S = S_0$; Ψ intact; $\Gamma_{\text{eff}} = 0$; reformation sequence active
Translation Function	$T(S, \Psi, \Gamma)$	$\Sigma(S) \geq \Sigma_{\text{min}}$ (0.40) — coherence enabling condition satisfied	Filters S through Ψ identity attractor projection; applies Γ modulation to produce V	V defined with Trajectory, Force, Aim, Continuity, Amplitude components
Feedback Function	$F(V, B_{\text{env}}, \Gamma)$	B(t) expressed	Computes $\partial S / \partial t$ from	$S' = S + \Delta S \cdot dt$;

Operator	Symbol	Trigger Condition	Primary Operation	Postcondition
		and environmental response B_env received	current V, environmental response, and gradient field; applies to update S	updated stance configuration ready for next translation cycle
Permeability Operator	$P(S, \Gamma) \rightarrow \partial S / \partial \Gamma$	Continuously active during all phases where $\Gamma_{\text{eff}} > 0$	Computes the rate of stance change with respect to gradient field pressure as a dynamic function of both S and Γ	Returns $\partial S / \partial \Gamma$ — used by feedback function for stance update computation
Identity Alignment Score	$A(S, \Psi) = \ \text{proj}_{\Psi}(S)\ / \ S\ $	Computed at every stance evaluation cycle; required by T, F, and CP	Projects S onto Ψ manifold; normalizes by magnitude of S to produce alignment score $\in [0, 1]$	Returns scalar $A \in [0, 1]$; used in translation filtering, continuity computation, and drift monitoring
Coherence Signature	$\Sigma(S) = (1/n(n-1)) \cdot \sum_{i \neq j} \langle S_i, S_j \rangle$	Continuously computed across all stance dimensions; primary health metric	Computes normalized mean pairwise inner product across all stance dimension vectors	Returns scalar $\Sigma \in [0, 1]$; governs vector production enabling condition and collapse risk assessment
Collapse Prediction	$CP(S, \Gamma, t) = w_1 \cdot (1 - \Sigma) + w_2 \cdot (1 - A) + w_3 \cdot (\Gamma \cdot P) + w_4 \cdot CI$	Continuously computed by HRS; evaluated at each monitoring cycle	Integrates four leading collapse indicators into a single composite risk score	Returns scalar $CP \in [0, 1]$; >0.80 triggers RETURN operator pre-staging by HRS
Confidence Envelope	$CE(t) = [\Sigma \cdot A] / (1 + \Delta \Gamma / \Delta t)$	Computed for all behavioral predictions; required by Section 10	Computes the ratio of coherent identity alignment to	Returns scalar $CE \in [0, 1]$; attached to all behavioral

Operator	Symbol	Trigger Condition	Primary Operation	Postcondition
		prediction algorithms	gradient field volatility; bounds all behavioral predictions	prediction outputs as confidence bound
Field Integration	$\oplus$	Applied when multiple gradient field components are superimposed	Vector sum of component gradient fields: $\Gamma = \sum_i \gamma_i$	Composite Γ ready for modulation application in Translation Function
Field Collapse	$\otimes$	Applied during S_x to collapse stance to ground state	Reduces all active stance field values to $\emptyset$ while preserving Ψ	$S = \emptyset$; Ψ intact; ready for RETURN sequence execution

Appendix B — Symbol Glossary

Complete alphabetical glossary of all symbols, variables, and operators used throughout the Behavioral Theory Specification. Symbols are listed alphabetically by the Roman character component; Greek symbols follow the Roman alphabet section.

Symbol	Name	Type	Defined In
$A(S, \Psi)$	Identity Alignment Score	Scalar $\in [0, 1]$	Section 3.4
A_{crit}	Identity Alignment Critical Threshold	Scalar (default: 0.15)	Section 3.6
$ B $	Behavioral Amplitude	Scalar $\in \mathbb{R}^+$	Section 4.6
$B(t)$	Behavior at time t	Point in behavioral space	Section 2.3

Symbol	Name	Type	Defined In
B_env	Environmental Behavioral Response	Vector	Section 6.1
BSP	BTS Summary Packet	Integration interface object	Section 11.6
BTS	Behavioral Theory Specification	Document identifier	Section 1.3
C	Continuity	Scalar $\in \mathbb{R}^+$ (integral)	Section 4.5
CA	Collapse Alert	Integration interface signal	Sections 11.3, 11.5
CE(t)	Confidence Envelope	Scalar $\in [0, 1]$	Section 10.5
CP(S, Γ , t)	Collapse Prediction Score	Scalar $\in [0, 1]$	Section 10.4
F	Force	Scalar $\in \mathbb{R}^+$	Section 4.3
F(V, B_env, Γ)	Feedback Function	Operator $\rightarrow \partial S / \partial t$	Section 6.1
FMF	Fractal Mechanics Foundation	Co-architecture document	Section 1.3
FSP	Fractal Scale Parameter	Integration interface object	Section 11.4
HRS	HERMES Runtime Specification	Co-architecture document	Section 1.3
IDR	Identity Drift Report	Integration interface object	Section 11.3
k	Amplification Coefficient	Scalar (environment-specific)	Section 9.3
n	Number of Stance Dimensions	Positive integer	Section 3.3
OER	Operator Execution Request	Integration interface signal	Section 11.5
P(S, Γ)	Permeability Operator	Operator $\rightarrow \partial S / \partial \Gamma$	Section 3.5
P_default	Default Permeability	Scalar	Section 7.8

Symbol	Name	Type	Defined In
	Value		
$R(\Psi)$	Resistance Coefficient	Scalar (Ψ -dependent)	Section 6.4
RDR	Recursion Depth Register	Integration interface object	Section 11.4
RSS	Runtime State Signal	Integration interface signal	Section 11.5
S	Stance	Internal cognitive state vector	Section 2.1
S_0	Proto-Stance	Ground state	Section 7.2
S_1	First Coherence	Initial stable state	Section 7.3
S_1'	Reformation	Post-collapse coherence state	Section 7.6
S_∞	Stable Stance	Asymptotic attractor state	Section 7.7
S_x	Collapse State	Dissolution state	Section 7.4
$T(S, \Psi, \Gamma)$	Translation Function	Operator $\rightarrow V$	Section 5.1
$\hat{t}$	Trajectory Unit Vector	Unit vector in behavioral space	Section 4.2
t	Time	Real-valued time parameter	Throughout
UCA	Unified Cognitive Architecture	Parent architecture	Section 1.3
UIA	Unified Identity Architecture	Co-architecture document	Section 1.3
UTMD	Unified Theory Master Document	Capstone co-architecture	Section 1.3
V	Vector	External behavioral trajectory	Section 2.2
w_1, w_2, w_3, w_4	Collapse Prediction Weights	Scalars summing to 1	Section 10.4
∂	Partial Derivative	Differential operator	Section 1.4

Symbol	Name	Type	Defined In
$\rightarrow$	Causal Direction	Relational operator	Section 1.4
$\oplus$	Field Integration	Operator	Section 1.4
$\otimes$	Field Collapse	Operator	Section 1.4
$\oslash$	Identity Nullification	Pathological condition operator	Section 1.4
$\curvearrowright$	Recursive Feedback	Relational operator	Section 1.4
Γ	Gradient Field	External pressure field	Section 2.6
$\Gamma_{\text{effective}}$	Effective Gradient Field	Γ as seen by agent (suppressed to 0 during S_0 reset)	Section 7.5
Υ_i	Gradient Component Field i	Component of Γ	Section 9.2
ΔS	Stance Drift Integral	Cumulative stance change vector	Section 6.5
θ	Angle between Γ and V	Scalar $\in [0^\circ, 180^\circ]$	Section 9.3
Ψ	Identity Field	Persistent attractor manifold	Section 2.5
ΨMN	Ψ Modification Notification	Integration interface signal	Section 11.3
ΨPC	Ψ Preservation Command	Integration interface signal	Section 11.3
ΨSP	Ψ State Packet	Integration interface object	Section 11.3
$\Sigma(S)$	Coherence Signature	Scalar $\in [0, 1]$	Section 2.4
Σ_{crit}	Coherence Critical Threshold (collapse)	Scalar (default: 0.20)	Section 3.6
Σ_{min}	Coherence Minimum Threshold (production)	Scalar (default: 0.40)	Section 5.3

Appendix C — Collapse-Reformation Quick Reference

This appendix provides a single-page operational summary of the complete collapse-reformation cycle, including all threshold values, decision criteria, state definitions, and operator references. This is the primary operational reference for HRS implementation and runtime monitoring of the collapse-reformation cycle.

C.1 Threshold Values (Default Configuration)

Threshold	Symbol	Default Value	Meaning	Action When Breached
Coherence Production Minimum	$\Sigma_{\min}$	0.40	Minimum Σ for V production	V undefined; no behavioral output
Coherence Critical Threshold	Σ_{crit}	0.20	Collapse Condition 1 trigger	RETURN operator triggered
Identity Alignment Critical	A_{crit}	0.15	Collapse Condition 2 trigger	RETURN operator triggered
Collapse Prediction — Imminent	CP_{imm}	0.80	Collapse imminent threshold	HRS initiates RETURN pre-staging
Collapse Prediction — High Risk	CP_{high}	0.60	High risk threshold	Activate monitoring; prepare re-centering
Drift Alert Threshold	ΔA_{alert}	-0.10 / episode	Identity drift monitoring flag	Increased $A(S, \Psi)$ monitoring frequency
Drift Critical Threshold	ΔA_{crit}	-0.25 / episode	Re-centering protocol trigger	Initiate Mechanism 2 or 3 re-centering

Threshold	Symbol	Default Value	Meaning	Action When Breached
Confidence — High	CE_high	0.70	High-confidence prediction	Predictions reliable; act on outputs
Confidence — Speculative	CE_spec	0.20	Speculative prediction threshold	Flag predictions; switch to monitoring mode

C.2 Collapse Condition Decision Matrix

Condition	Label	Formal Criterion	Primary Cause	Collapse Type
1	Coherence Collapse	$\Sigma(S) < 0.20$	Accumulated contradiction; sustained pre-collapse operation	Gradual — predictable
2	Identity Ejection	$A(S, \Psi) < 0.15$	Severe identity drift; Ψ -foreign stance configuration	Gradual — drift-driven
3	Gradient Overload	$ \Gamma > P^{-1}(S)$	Boundary breach; extreme environmental pressure	Acute — event-driven
4	Recursive Contradiction	Mutually nullifying sub-dimensions; oscillation without resolution	Internal logical conflict in stance configuration	Internal — self-generated

C.3 State Definitions Quick Reference

State	Symbol	Σ Range	A(S, Ψ) Range	V Status	Behavioral Capacity
Proto-Stance	S_0	Undefined	Undefined	Undefined	None — formation only
First Coherence	S_1	0.40–0.55 (marginal)	Low–Moderate	Defined (minimal)	Exploratory; low force; moderate continuity
Moderate Operation	S_n	0.40–0.75	Moderate	Defined	Functional; variable force; moderate continuity
High Coherence	S_n	>0.75	High	Defined (stable)	High force; high continuity; predictable
Collapse State	S_x	<0.20	<0.15	Suspended	None — RETURN in progress
Reformation	S_1'	0.40–0.55 (reformed)	Moderate–High (Ψ -guided)	Defined (reformed)	Exploratory; identity-shaped; improving
Stable Stance	S_∞	→ Maximum	→ 1.0	Defined (optimal)	Maximum amplitude; optimal continuity; resonance-capable

C.4 RETURN Operator Sequence — Compressed Reference

RETURN OPERATOR — RAPID EXECUTION REFERENCE [RETURN: $S_x \rightarrow S_0$]
 PRE-CHECK: Confirm S_x | Confirm Ψ intact | Confirm Σ
 <
 0.20

STEP 1: Suspend V $V = \text{undefined}$
 STEP 2: Log collapse context $\{\Gamma, \text{last_}\Sigma, \text{last_}A, \text{cond}\}$ STEP 3: Archive
 S $S \rightarrow S_{\text{archive}}$ STEP 4: Clear
 S $S = \emptyset$ STEP 5: Confirm Ψ
 verify integrity; flag if $\emptyset$ STEP 6: Reset P $P =$
 P_{default} STEP 7: Suppress Γ $\Gamma_{\text{eff}} = 0$ STEP 8:
 Init S_0 reformation sequence active STEP 9: Form S_1
 ' guided by Ψ attractor landscape STEP 10: Restore Γ
 when Σ

 $>$

 $\Sigma_{\text{min}} \rightarrow \Gamma_{\text{eff}} = \Gamma$

POST-CHECK: S_0 active | Ψ intact | $\Gamma_{\text{eff}} = 0$ | Reformation $\heartsuit$
 COMPLEXITY: $O(n)$ — n = stance dimensions at time of collapse
 REVERSIBILITY: NOT REVERSIBLE — S_{archive} is record only